Learning Dexterous Manipulation: Hands, Simulators, Training, and Evaluation

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***Abstract*—A dexterous hand can change an object's pose without setting it down. This survey covers 113 learning-based methods that attempt it, the hands they run on, the simulators they train in, and how they report results. The field has largely converged on one recipe: reinforcement learning on privileged state in a GPU simulator, distilled into a vision-only student for transfer to hardware. Almost half the corpus trains that way, nearly all of it in one simulator, whose own successors are barely used. A second and faster-growing branch learns from teleoperated demonstration instead, through vision-language-action models, diffusion policies, and systems whose contribution is the collection rig itself. Published work has tripled since 2022. Grasping remains the largest task family and bimanual coordination the fastest-rising, and almost every experiment runs on one of four hands. What the survey adds comes from reading released code and vendor specifications beside the papers. Eight hands that can be bought or built from published designs appear in no method paper here. Where the depth a hand sinks into the object it holds is reported at all, it is not measured in the physics that produced the motion. Section VII proposes a protocol for the unreported quantities; the body gives the counts behind every claim above.**

***Index Terms*—Dexterous manipulation, multi-fingered hands, bimanual manipulation, reinforcement learning, imitation learning, physics simulation, contact modeling, benchmarking, reproducibility.**

I. Introduction

A hand is dexterous when it can change an object's pose without putting the object down. That property, and not the finger count, is what separates a hand from a gripper. Bicchi [1] draws the same line in Sec. 1.1 between restraining an object and manipulating it with the fingers: restraint is a static question about whether the contacts prevent motion, while in-hand manipulation is a dynamic one about whether contacts can be broken and remade while the object stays held. A parallel-jaw gripper can hold almost anything and reorient almost nothing.

The classical theory of grasping is about holding, and it is good at it. Given a set of contact points and an assumption about friction, it can decide whether an object is immobilized, and what force each finger has to apply to keep it that way. What it cannot do is say what the fingers should do next. Contact is what makes that hard: a finger is either touching or it is not, and it can push but not pull, so the equations of motion change form every time a contact is made or broken. The standard treatment says as much about its own results, that immobilizing an object "does not guarantee stability", and that "the nonsmooth nature of grasp dynamics, because of the unilateral constraints on displacements and forces, has made a thorough analysis very difficult" ([1, Sec. 1.5 and 1.8]). The tradition produced tests. You can check a candidate grasp against them; you cannot run them on a robot.

Learning changed what a practitioner has to produce. The old deliverable was a proof that a configuration holds. The new one is a policy that scores well over many simulated attempts, which means the simulator now stands in for the analysis and the score stands in for the proof. That is why the field moved, and it is also why this survey is organized as it is. It shifts two burdens onto the author. The simulator has to be faithful where fidelity matters. The measurement has to mean something. Much of what follows asks how well this literature carries those two.

One recipe now carries most of the field. Train a policy with reinforcement learning on privileged state, the object pose, velocities and physical parameters the robot cannot sense, in a GPU simulator running thousands of environments in parallel; distill that teacher into a student that sees only what a camera and joint encoders provide; transfer the student. Fifty-four of the 113 method papers here train this way, most of them in one simulator, and about half of those add the distillation stage. In-hand cube reorientation is the recipe's benchmark, inherited from [2] and still the task a new method is expected to show.

A second branch is growing underneath it and inverts the dependency. Instead of a simulator and a reward it takes human demonstrations through a teleoperation rig and fits a policy directly, as a vision-language-action model, a diffusion policy, or a data-collection rig whose design is itself the contribution. Section V counts them, and Fig. 1 shows how fast they have arrived. It is the newer literature, most of it from the last two years, and it trades the reward-design problem for a data-collection problem. Which of the two branches ends up feeding the other is not settled here, but the growth is the clearest trend in the corpus.

Both branches run on remarkably little hardware. Four hands carry almost every experiment in this corpus, led by the Allegro at 35 of the 113 papers; Fig. 3 gives the rest of the distribution. Eight further hands can be bought today or built from published designs and appear in the experiments of none of them. The hardware exists and is purchasable, so this is

Correspondence: luai_abuelsamen@berkeley.edu, a personal address rather than an institutional one: this work was done independently, with no department, grant or employer behind it. Every claim in this paper traces to a structured note taken from a parsed source, and every count is recomputed from those notes at build time. The corpus, the notes and the build that produce them are available from the author on request, and will be deposited with a persistent identifier.

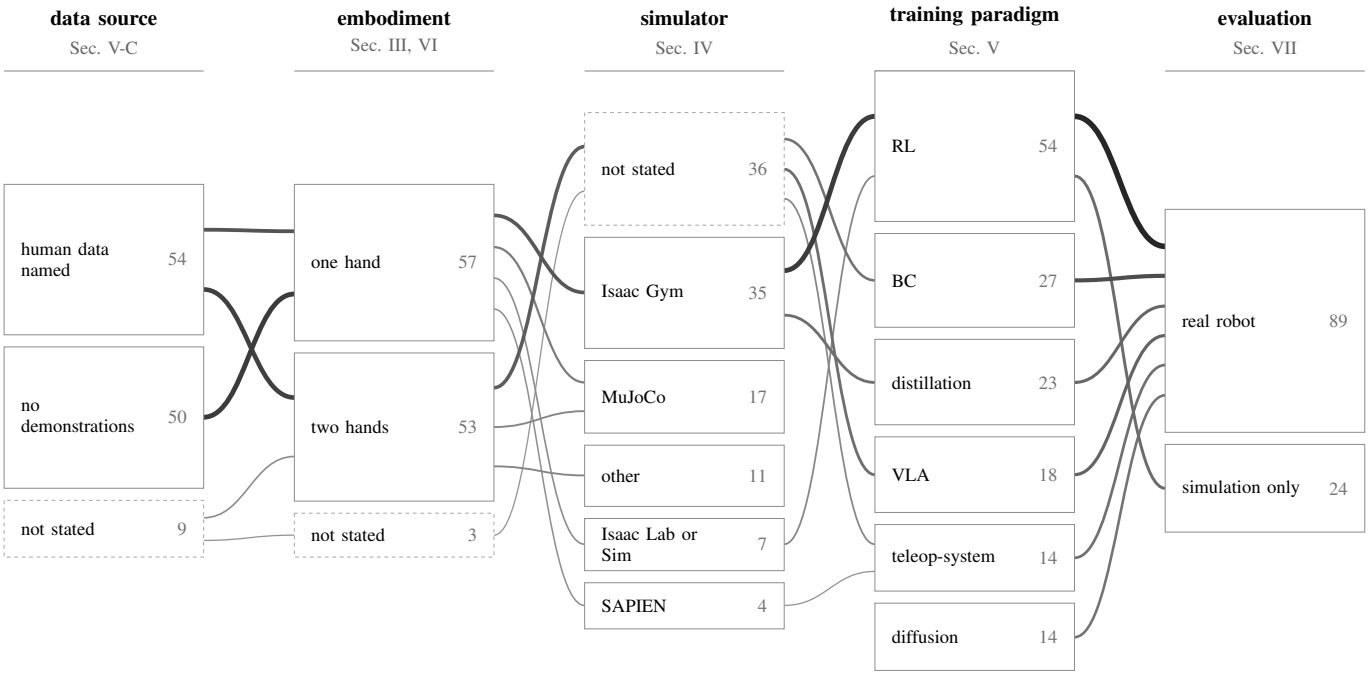


Fig. 1: The field on one page, and the map of this survey: nodes sized by corpus papers, edges drawn for the routes that carry at least two fifths of the smaller endpoint, so the dominant path reads as a path, each column head naming the section that covers it.

not a manufacturing lag, and Section III takes it up: a field whose results rest on four platforms is more fragile than its publication volume suggests.

What this literature cannot currently do is compare its own results. Nearly every paper states a success criterion, but they are not the same criterion, and only 70 of the 113 say how many trials a reported rate rests on. The criteria run from never dropping the object to reaching a pose within a tolerance each paper picks for itself. Two success rates from two papers on nominally the same task are usually not measuring the same event, and none of the surveys preceding this one supplies a protocol that would make them comparable. Section VII derives one, with the trial counts a stated confidence interval actually requires.

Every number in this survey is traceable to a source, and every count in it is recomputed from those sources rather than typed in. Appendix A says how the corpus was built, what the method cannot do, and why each coverage figure here should be read as a floor rather than a rate.

One part of that method is unusual enough to flag now. Where a paper published a repository, its reward was read from the repository as well as from the paper, so the two can be compared. Reading an implementation against its own description is established practice outside robotics, in repeatability and reproducibility studies [3], [4], in audits of how implementation detail moves reinforcement-learning results [5], in reward-design review [6], in benchmark re-releases [7], and most directly in recent work on paper–code alignment [8], [9]. We are aware of none that does it for a robotics field's released reward functions. Section V-F reports what this one found.

The scope is single-hand and bimanual multi-fingered manipulation under learned control. Parallel-jaw work enters as a comparison and not as a subject. Two limits apply to every number here. The corpus is of the learned era, which is a selection and not a verdict on the analytic tradition; and where this survey failed to extract a value, the counts look the same as if the source had never stated one.

A. How to read this survey

Fig. 1 puts the field on one page, and the sections work outward from the task. Section II sets out the task families, what makes each one hard, and the handful of terms this survey uses precisely. Section III covers the hands and their sensing, Section IV the simulators and what they expose about contact, Section V how policies are actually trained, Section VI what changes when there are two hands, and Section VII how any of it is evaluated. Section VIII says what the corpus supports and what it does not. The appendices hold the method and the full tabulation: A the method and its limits, B the hardware and engine tables, C every paper-against-code comparison, D the surveys that precede this one, E the derivation behind the protocol.

If you are new to the field, read Section II for the vocabulary, then the first half of Section V for the two ways a dexterous policy gets trained, then Section III to see what hardware any of it runs on. Those three give the shape of the field in about half an hour. Section VII is the one to read before designing an experiment, because it is about what the field's numbers do and do not mean.

If you already work here, the parts likely to tell you something you did not know are Section IV-B on what simulators expose about contact, Section V-F on papers against their own released code, Section VI on bimanual work as its own problem rather than two single hands, and Section VII-C on what a success rate at twenty trials can support.

If you want the data rather than the prose, every table here is generated from the corpus and every number in the text is recomputed from it at build time: Tables IV and V for the hands, Table VI for the simulators, Table II for the paper-against-code comparisons and Table III for the proposed protocol. The paper-by-paper tabulation is in the corpus rather than in these pages, and Appendix A says where.

Three things to carry while reading. The corpus is of the learned era, which is a selection and not a verdict on the analytic tradition that preceded it. Every coverage figure is a floor: when a paper is silent and when this survey failed to extract a value, the counts look the same, and Appendix A says which direction that bias runs. And a success rate in this literature is rarely comparable across two papers, for reasons Section VII sets out, so prefer the sections that explain mechanisms to any ranking assembled from the numbers.

II. A TAXONOMY OF THE PROBLEM

A. Task families

Dexterous manipulation is not one task, and what separates its tasks is what makes some of them hard: what is held, what moves, which way gravity points, and what counts as failure. Six families cover most of the corpus, and the last paragraph here says what they miss. Grasping is the largest, with 58 of the 113 papers carrying the label; the labels are not exclusive, so one paper can sit in several families. Success is a lift that survives a hold, and the thresholds differ by more than an order of magnitude: one paper requires the object "held 10 cm above the table for 20 s" [10], another "held at least 0.5 s in simulation" [11]. What makes grasping hard at scale is the continuum of starting configurations. UniDexGrasp++ [12] states it in Sec. 4.3, that "we are dealing with an infinite number of tasks considering the initial object pose can change continuously".

Functional and tool use is second with 45 papers. It is the family where a stable grasp can still be the wrong answer. Dexterous Functional Grasping [13] gives the case in Sec. 2.1, that "grabbing a hammer from the head or handle are both equally valid ways of using it", and only one of the two lets the tool be used. Success is defined against the tool's function, and the field has no shared way to state it.

In-hand reorientation has 31 papers and the most settled criteria, because the community inherited one number: OpenAI declared the goal achieved below 0.4 rad of orientation error [2], and its successors keep that tolerance or tighten it [14], [15]. A shared tolerance is not a shared protocol, because the stopping rule differs. Visual Dexterity [16] measures error "when the controller predicts it has reached the goal and stops", which makes the policy a judge of its own trial.

Tracking a human reference has 12 papers and inverts the problem. The target is a trajectory rather than an endpoint, so success is a per-frame error band, and the bands are compound: one method requires rotation, translation, joint and fingertip error all to stay inside separate thresholds at once [17]. The difficulty is that the reference came from a human hand and was never dynamically feasible for the robot. DexMachina [18] warns in App. B.4 that scoring by timesteps inside the thresholds makes "the results highly sensitive to the threshold values".

Bimanual coordination has 43 papers and handover has 10. Handover is the smallest family and the one whose failure has a single moment, because the giver must release only after the receiver has the object. Dynamic Handover [19] reports a hit rate above its success rate and blames the gap in Sec. 5.4 on "occasional challenges encountered during the grasping phase of the catcher". Not every paper here learns both sides. In [20] only the receiver is learned and the giver is a scripted arm.

Those six families do not cover the corpus. Twenty-four of the 113 papers carry a label from outside them and eight carry no label from the six at all. Twenty fall in a catch-all class, mostly generalist policies evaluated on a task suite rather than on a dexterous task family, among them [21], [22], [23], [24] and [25]. Three adjacent families are named here instead of being folded into the six. Locomanipulation is [26], [27] and [28], where the base is not fixed and the gravity argument below changes character. Piano playing is [29], [30] and [31], discussed in Section VI, where success is a per-timestep F1 against a MIDI score rather than an object pose. Deformable manipulation is [32], whose object carries its own state and its own physics. Ferrari and Canny's grasp-quality measure [33] carries no task label, because it is a measure rather than a task. Table I's six families are the ones with enough papers to compare, and not a partition of the corpus.

B. Sources of difficulty

Contact is non-smooth, and that is a property of the problem rather than of any solver. Bicchi [1] states in Sec. 1.2 that contact constraints are unilateral, and that losing a contact "involves an abrupt change of the structure of the model under consideration". Its Sec. 1.4 gives the sharper version, that a rod sliding on rough ground has configurations with no consistent solution and configurations with more than one. Simulators inherit the difficulty. The re-implementation study [34] concludes in Sec. V that "there is no fully satisfactory approach at the moment, as all existing solutions compromise either accuracy, robustness, or efficiency". Pang et al. [35] treat non-smoothness as the thing to remove, curating every contact pair in Sec. III-D so that contact points and normals change smoothly.

The hand has more actuators than the object has degrees of freedom and is still short of authority over it. The object moves only through contacts, and the contacts are what cannot be commanded. Bicchi [1] states in Sec. 1.4 that in multifingered grasps "the number of independent contact forces is much larger than the number of actuators. Thus, from a controllability standpoint, not all the contact forces are controllable".

Vision is occluded by the hand doing the work. Bai et al. [36] name it in Sec. 4.3, that "occlusion hampers object tracking". DexPoint [37] keeps vision and adds imagined hand points to the point cloud. Yin et al. [38] remove vision entirely and rotate objects from 16 binary touch sensors over the palm, links and fingertips.

Gravity direction changes the task rather than scaling it. Visual Dexterity [16] states that reorientation with the hand

below the object "is much easier", because "with a downward-facing hand, the hand must manipulate the object while simultaneously counteracting gravity". AnyRotate [39] makes the direction a randomised variable by "randomly initializing hand orientations between episodes", and its Sec. 5.3 measures the cost, with performance dropping progressively from palm up and palm down, through base up and base down, to thumb up and thumb down.

Three counts describe two hands and they measure different things. Fifty-three of the 113 papers record two hands on the robot, which is all the corpus's two-hand flag claims. Forty-three carry the task label for bimanual coordination, the narrower claim that coordinating the hands is the task. Section VI narrows again, to the 28 papers whose notes place a learned closed-loop controller on two multi-fingered hands, and its opening paragraph names every exclusion that takes the 53 down to the 28. That 28 is the denominator for every architecture count in this survey. Every bimanual claim here names which of the three it uses.

Four things genuinely change when the second hand arrives. Contact stays non-smooth, occlusion stays, and gravity stays the same problem.

Role asymmetry is one. AsymDex [40] assigns a dominant hand with full finger and wrist control and a facilitating hand with 6-DoF base pose only, so that "the facilitating hand repositions and reorients one object, while the dominant hand performs complex manipulations".

Two hands on one object also close a kinematic chain through the object. In [1] Sec. 1.4 the hand and object dynamics are separate and are "linked through the n rigid-body contact constraints". A second hand adds a second constraint set on the same object rather than a second independent problem. That is where physical plausibility fails first. BimanGrasp [41] is the one paper here that measures it, rejecting a synthesised grasp when "total penetrations exceeded 1.5 mm" and reporting in Sec. IV-C that "penetration remains the primary cause of grasp failure".

Two-arm collision does not exist for one hand at all. DyDexHandover [42] resets an episode when "any unintended arm contact with the environment or self-collisions" occurs. Bunny-VisionPro [43] pays for it in the controller, where adding a sphere-approximated self-collision cost raises motion-control time from 0.74 ms to 7.85 ms. Bidex [44] solves it in hardware, mounting the hands so the human arm and the teacher arm "are perpendicular to each other and do not collide". DexMimicGen [45] states plainly that it does not handle inter-arm collision at all.

The action space doubles as well. Bi-DexHands [46] runs 20 tasks on two Shadow Hands and finds single-agent PPO beating multi-agent RL on most of them, offering in Sec. 5.2 that "PPO algorithm is able to use all observations for training the policy, while MARL can only use partial observations". AsymDex [40] attacks the dimensionality from the other side, halving the observation and action dimension through its role split and a frame relative to the facilitating hand's object.

Two cells say the field has no agreed criterion, and both are in families where a second body is involved. A criterion naming a distance, a duration and a trial count can be re-run by someone else. A rubric cannot, and Section VII takes up what follows from that.

C. Terms used throughout

Six terms recur in every section that follows, and each is used here in one sense only.

Method paper. A paper whose own experiments train or evaluate a dexterous policy. It is the unit every count is taken over: 113 of the corpus's 219 entries are method papers, the rest being hands, simulators, datasets, benchmarks, surveys, tactile sensors and evaluation protocols.

Privileged state. Information a simulator can supply and a robot cannot: object pose and velocity, contact forces, mass, friction. A policy trained on it cannot be deployed as it stands, which is what makes the next term necessary.

Teacher–student distillation. Training one policy on privileged state, then fitting a second that sees only deployable observations, usually camera images and joint encoders, to reproduce the first. Section V finds 23 of the 113 methods do this, and it is the second half of the recipe Section I describes.

Reference and rollout. A reference is a pose or a trajectory scored before anything executes: a retargeted human motion, a synthesized grasp, a planned path. A rollout is what a trained policy actually did. The distinction matters because a quantity can be measured carefully at the reference and never at the rollout, which Section VII finds is the case for interpenetration.

Interpenetration. The distance by which two bodies that should be touching instead overlap, a consequence of solving contact constraints approximately rather than exactly. Section IV covers what each engine exposes; the depth is a property of the compliance someone chose, not of the engine.

Success criterion and per-cell trial count. The criterion is the predicate a paper uses to call an episode a success: a pose within a tolerance, an object never dropped, a task completed. The per-cell count is how many trials one reported rate is computed from. Section VII finds 98 of them state a criterion and no two state the same one, and that 70 state a count.

D. What has changed since 2022

The corpus roughly tripled between 2022 and 2024, and its composition changed while it grew. Three shifts are visible in the record, and none of them is small.

Sorting each method by whether it trains in simulation, through reinforcement learning or distillation, or from human demonstration, through behavior cloning, a vision-language-action model, a diffusion or flow policy, or a collection rig, the simulator branch led 13 papers to 4 in 2023 and trailed 15 to 18 in 2024. It has trailed since: demonstration-trained work is 22 of the 34 papers from 2025 and 8 of the 13 so far from 2026. The recipe Section I describes is this field's mature pipeline rather than its growth area, and a reader who knows only that recipe is reading the field as it was in 2023.

It was 8 of the 11 papers in 2022 and 3 of the 34 in 2025. The absolute count falls while the corpus triples, which is not a shift in emphasis but a change in what the field treats as worth publishing. Bimanual coordination moved the other way over the same span, from one paper to 16, and functional and tool

TABLE I: Task families against the properties that define success.

task family	what is held	what moves	what gravity does	what counts as failure	shortest honest success criterion
In-hand reorientation	one rigid object, held by the fingers for the whole episode	the object's orientation in the palm frame, while contacts break and remake	a disturbance whose direction is set by hand orientation, and a load to counteract with the palm down, [16]	the object leaves the hand, e.g. the fall reset at goal distance 0.24 m in [14]	orientation error below a stated tolerance under a stated stopping rule, 0.4 rad in [2] and [14], 0.1 rad in [15]
Grasping	nothing at the start, the object rests on a support	the hand closes, then the object is lifted clear	the test itself, since the grasp exists to resist it	no lift, or a drop inside the hold window	a stated lift height held for a stated duration, 10 cm for 20 s in [10], 10 cm to episode end in [47], 0.5 s in [11]
Functional and tool use	a tool, held in the pose its function requires rather than the most stable one	the tool's own joint, or the tool acting on a second object	secondary, the function dominates	a stable grasp in the wrong place, [13]	no agreed criterion. Criteria range from never dropping the object, [13], to a four-point rubric, [48]. The one mechanical threshold is holding the tool joint past a commanded angle for the episode, [49]
Tracking a human reference	whatever the human held in the captured clip	hand and object follow a recorded trajectory under physics	full, and the reference never obeyed it, so the reference is not dynamically feasible	per-frame error leaves the band, or the object drops	all stated per-frame errors below threshold for the whole clip, 30 deg and 3 cm and 8 cm joint and 6 cm fingertip in [17]. Thresholds are not comparable across papers, and DexMachina [18] calls the results "highly sensitive to the threshold values"
Bimanual coordination	one object by two hands, or one object per hand	both hands, the object, and the relative pose of the hands	can be carried by one hand while the other acts, [40]	the object drops, the arms collide, or the hands work against each other	no agreed criterion. Task-specific thresholds dominate, e.g. 0.035 m block-to-cup distance in [40]. The one criterion with a physical-plausibility gate is 2.0 s of hold with total penetration under 1.5 mm, [41]
Handover and in-hand transfer	by the giver at the start, by the receiver at the end	the object crosses between hands, released only after it is regained	unopposed during the exchange, and total in a thrown transfer, [19]	a drop at the moment of release, and the catcher's grasp phase, [19] Sec. 5.4	the receiver holds the object and moves it clear, more than 10 cm from the first hand in [50], or holds it to episode end in [42]

use from 2 to 15. The cube-rotation benchmark that Section V shows the reward literature organized itself around is the task the newer literature has left.

No paper earlier than 2025 uses Isaac Lab or Isaac Sim at all. Three do in 2025 and four in the partial 2026, against nine and two respectively on Isaac Gym, so on small numbers the successors already carry more of the most recent year than the original does. A survey assembled twelve months earlier would have recorded a field that had not moved, which is worth stating plainly: the concentration this survey reports on one simulator is a fact about a corpus with a cutoff, not a permanent property of the field.

III. Hands and who makes them

A. Design axes

A policy is only as dexterous as the hand it runs on, and the hands available divide sharply: a few a laboratory can buy, a few more it can build from published designs, and a growing number that exist only in an announcement. What can be said about them is less than it looks. No degree-of-freedom, force, weight or price figure in Table IV or Table V, both set in Appendix B, was measured by anyone outside the maker. Six entries are the exception, from peer-reviewed papers with stated protocols: ILDA, Pisa/IIT, ORCA, RUKA, LEAP and BiDexHand. Every other claim is dated at its source. Table V prints the date beside each announced hand, and the repository tabulation carries a claim-date column for the hands that can be bought, running from Shadow's specification of 4 December 2024 to vendor pages fetched on 17 September 2026.

The degree-of-freedom count is the first number a vendor states and the least comparable one, because three different quantities get printed under that heading: joints, actuated joints and actuators. Shadow's own specification separates them, reading "20 actuated DOF and a further 4 under-actuated movements for a total of 24 joints" [51]; Inspire's page gives a DoF figure and a joint count that differ by a factor of two [52]; a tendon hand can list more actuators than it has kinematic degrees of freedom [53]. Table IV therefore prints joints and actuated DoF in separate columns, with actuators kept in a third in the repository tabulation, and the gap between them is the informative number.

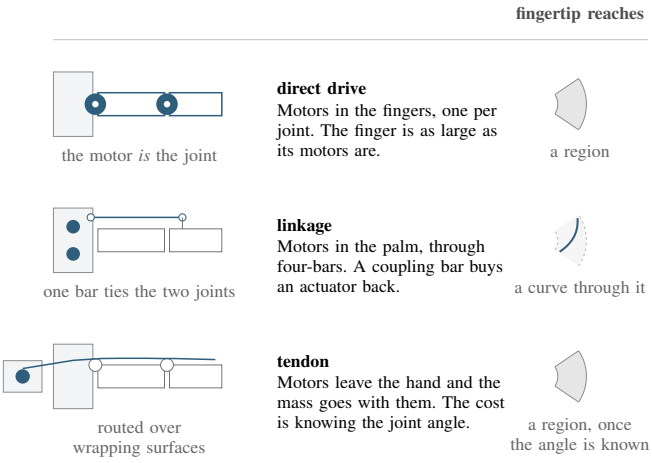


Fig. 2: Three answers to where the motor goes, and what each one costs. A coupling bar saves an actuator by tying two joints, so the fingertip reaches a curve through the reachable region rather than the region itself.

The actuation ratio is the real decision, and Fig. 2 sets the three answers side by side. The Pisa/IIT SoftHand drives nineteen joints from one motor, and its authors state the cost

plainly: “No in-hand dexterous manipulation is required for this prototype” [54]. Tendon drive moves the motors off the fingers and the mass follows them, so a hand with every motor in its palm can weigh a quarter of a tendon hand with a forearm [51], [55]. What tendons cost is state estimation. The Faive Hand estimated joint angles from tendon length through a Kalman filter and its cube reorientation failed on hardware, which its authors blamed on “poor joint angle measurement from the EKFs, especially when there is contact” [56]. Ruka-v2 adds detachable encoders to measure that problem rather than close the loop on it, and they put its open-loop joint-to-motor map at several degrees of error [57].

Direct drive trades torque density for transparency: LEAP’s Dynamixel joints resist 19.5 N in a pull-out test against the Allegro’s 8.5 N [58]. Linkage drive is the argument against the forearm, putting motors, drivers and ball screws inside the palm and still reaching tens of newtons at the fingertip [55]. Linkages also buy joint coupling for free, as in BiDexHand’s four-bar that drives each distal joint off the one behind it, though that hand’s paper and its own README disagree about how many joints and servos it has [59].

Fingertip force is the number a hand buyer reads first, and across these hands it is six different measurements wearing one name. Pull-out resistance, a pinch between two fingers, fingertip normal force under a standard indenter, a calibrated multi-trial measurement, a peak vendor claim and two unlabelled columns on a spec sheet all appear as “fingertip force”. One vendor’s figure turns out to be the newton equivalent of a payload at a fixed motor current, which is a different quantity again [60]. Table IV therefore prints the protocol beside every force it reports and leaves the cell empty where the protocol cannot be recovered. Payload is no more comparable: nine hands state one and no two state it under the same test, so the repository tabulation carries it with its protocol and this table does not print it at all.

Compliance is repairability seen twice over, and both are mechanical fuses: the Pisa/IIT SoftHand’s rolling-contact joints return to assembly after over-extension and ORCA’s “poppable” pin joints dislocate rather than break [54], [60]. ORCA also reports tendon drive as maintenance: “Prolonged use requires manual re-tensioning to maintain performance”. What fails is rarely the motors. Its durability run found silicone skin degrading on two fingertips after about 2,000 to 4,000 grasp cycles and sensor wires snapping on three after about 4,500 to 7,000. The sensing wore out an order of magnitude sooner than the hand.

Two properties matter before any of the above, and no vendor comparison carries them; the repository tabulation prints both. The first is the rate at which a control loop can actually be closed, which spans a factor of four across these hands and is not stated at all by some. One hand’s quoted rate turns out to be a serial query ceiling while its own transferred policy runs at 20 Hz, which answer different questions [58]. A reader deploying torque control is bitten by that first. The second is whether a URDF or MJCF exists and who ships it, which is the bridge to Section IV and to this survey’s recommendation: prefer a hand a simulator already carries. LEAP is the warning there, because its paper says the URDF is released and the released API repository contains none [58].

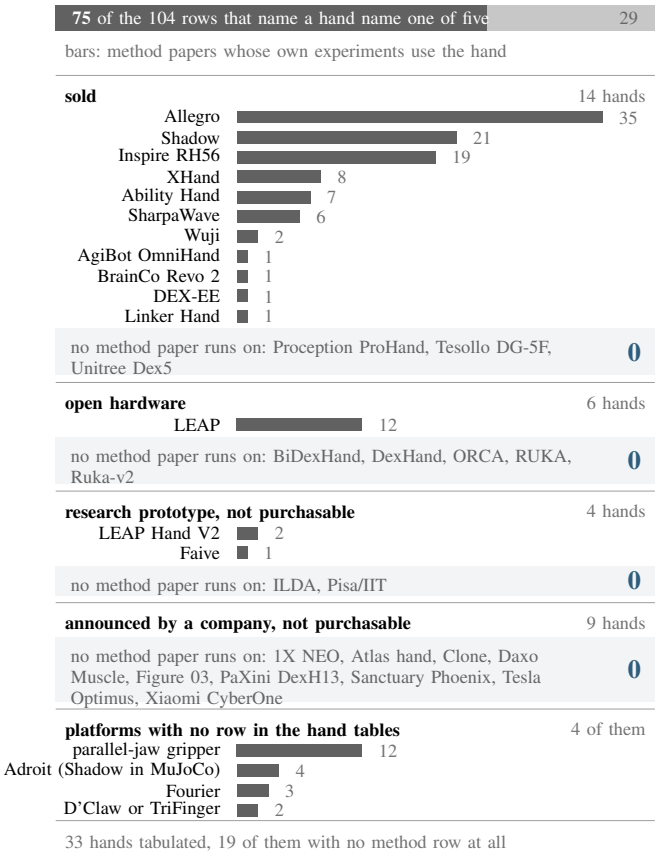


Fig. 3: Which hands the literature runs on, banded by how the hand can be got, with every tabulated hand shown and those no paper uses as named slots at zero. A paper naming two hands counts under both.

B. Hands in use, and what they cost

The two oldest designs in Table IV carry 49 of the 104 papers that name a hand at all, and seven papers use both. Fig. 3 counts, per hand, the papers whose own experiments use it, and the shape of that figure is the finding: four hands take three quarters of the corpus and everything else shares the remainder. What no paper states is how big any of them is. Neither hand table carries a dimension column, because no source behind them gives a hand’s length or its width: the specification a reader could hold against their own hand is the one these sources omit, while the degree-of-freedom count, which all of them give, is compatible with a hand of any size at all. Mass is the nearest thing on record and it is not the same quantity.

The Allegro’s position is the uncomfortable part. Its product page at allegrohand.com/v4 returned HTTP 404 while the Wonik wiki timed out [61]. Everything Table IV confirms about the most-used hand in the corpus comes from its ROS driver: 16 joints, four fingers, a torque interface, a 333 Hz CAN clock, 12 V, no tactile sensing. Its weight, joint torque, payload and price have no reachable source. The field’s most common platform cannot be specified from a page a reader can open.

Concentration would matter less if the hand did not move the result, and it does. On the same simulated cube rotation LEAP turns the cube nearly three times faster than the Allegro [58], and on the same three-trial pinch test RUKA reports almost twice the Allegro's force [62]. A method compared only on Allegro hardware is compared at one point in a space where a single axis moves the headline number two or three times over. The three newest vendor hands have arrived quickly and take a growing share, none of them before 2024, and eight hands that can be bought or built take no papers at all.

Six hands in Table IV are open hardware, and LEAP Hand V2 in Table V is a seventh. Five of the seven state a dollar cost: $2,000 for LEAP, $3,000 for LEAP Hand V2, $1,500 for Ruka-v2, $1,300 for RUKA and $300 for DexHand. ORCA states a material cost below 2,000 CHF that the price column leaves unconverted, and BiDexHand states none [60], [59]. The Faive Hand states neither a cost nor a license that could be read, so it is not counted as open hardware here [56]. Two of the stated bases need saying. DexHand's $300 is "additional total cost of components", excluding the printing and the wrist servos [63], and ORCA's own figure sits against Ruka-v2's table listing ORCA at about $3.5K [57].

The expensive end of that comparison is secondhand throughout: the six-figure prices for a Shadow Hand come from other people's comparison tables [62], [56], because the vendor's own page asks buyers to discuss pricing. So the collapse is real at the cheap end and inferred at the expensive one, and it has barely moved the literature. Twelve of the 104 hand-naming papers use an open-hardware hand, and all twelve of those are LEAP.

What the cheap hands give up is sensing. LEAP has none and names touch sensors as future work [58], RUKA states its design "lacks tactile sensing" [62], and BiDexHand's sources never mention touch [59]. ORCA is the exception, with binary FSR fingertips whose threshold was measured as low as 0.05 N on a fresh fingertip, against the sensor's rated 0.29 N, and 6.38 N on a degraded one [60].

C. Tactile sensing

DIGIT set the cost floor, stating a manufacturing cost of about 15 USD per sensor in batches of a thousand [64], which is where the price of touch sensing stopped being the obstacle. Durability was as much of the contribution: its gel barely degraded over an abrasion test that destroyed the two gels compared against it.

Digit 360 is the same lineage at a different operating point: millions of taxels, force resolution around a millinewton, and on-device processing that cuts event-to-action latency by most of an order of magnitude [65]. It is also not for sale, and no paper in this corpus names it.

XELA's uSkin is the non-camera alternative, and the one that bolts onto hands the field already uses: a curved fingertip kit for the Allegro and LEAP, with three-axis sensing points at gram-force resolution running at a few hundred hertz [66]. Table V gives the specification, including which figures belong to the product family rather than to the fingertip kit. One paper in this corpus uses it, and its self-supervised encoders beat matched end-to-end models when both see a fraction of the labels, though its own bead-maze policies never complete the maze on the real robot [67].

The usage number is the one to keep, and it needs its inclusion rule stated: a sensor physically on the hand, read by the deployed policy. Eight papers meet it: [39], [49], [68], [69], [50], [70], [71] and [38]. The rule excludes [72], whose 20 binary contacts are simulated on an Allegro without tactile hardware and whose released configuration disables tactile input; it also excludes [73] and [74] for the same reason. Sixty-three of the 113 method papers use the word tactile somewhere in their parsed text and 35 carry it in their structured note. Against the eight that meet the rule above, either number is the gap worth quoting.

Almost none of the eight uses a high-resolution sensor. Yin et al. [38] remove vision entirely and rotate objects from 16 binary touch sensors over the palm, links and fingertips, deployed zero-shot to a real Allegro, and Robot Synesthesia uses 16 force-sensing resistors read as binary [70]. Two papers exclude touch deliberately, HORA reporting rotation "even without the usage of vision and tactile sensing" [75], and ORCA's authors dropping it from their reinforcement learning "due to the additional complexity involved in accurately modeling them" [60]. Taxel counts have risen by three orders of magnitude while the policies consuming them have stayed at binary contact.

D. The catalog against the literature

Behind Table V's entries sit three kinds of evidence, and conflating them is how a DoF figure with no source ends up in a survey. Vendor prose carrying numbers is the strongest, as on 1X's page of 9 July 2026 [76]. Video is second and carries none. Press or bibliography assertion is third, as with Tesla's V3 hand, known here only through a paraphrase of patents because the USPTO PDF parsed empty [77]. Table V blanks the cells resting on the third class and footnotes who did the arithmetic.

A survey can go one step past recording that a claim is unverified, which is to say which claims are implausible on their face. One vendor's actuator count and mass work out to about six grams per actuator including structure, tendons, routing and skin [53]. Another's mass figure appears to exclude a half-kilowatt pump [78]. A third's degree-of-freedom figure sits on the same page as a finger count exactly half of it, so it is almost certainly the pair of hands [79]. A fourth, widely repeated for the Optimus hand, is one article's arithmetic rather than a vendor statement, and the previous generation's own figure was half of it [77].

Skepticism belongs to the evidence class, not to which table an entry lands in. The fully-actuated claims from the newest vendors are claims about unmeasured hardware, and they sit in Table IV beside the footnotes their own pages carry: "Specifications may vary between products" on one, a spec that "will continue to iterate" on a beta product on another [80], [81]. Even the best-specified vendor page in the corpus leaves its fingertip-force columns unlabelled.

The band at zero in Fig. 3 carries the finding. None of the nine company-announced hands in Table V appears in

a single paper whose own experiments use it. Tesla, Figure, 1X, Sanctuary, Boston Dynamics, Xiaomi, Clone, Daxo and PaXini account for zero of the 113 method papers' experiments. The nearest thing to a counterexample is [82], a Figure blog post claiming a 35-DoF whole-upper-body action space at 200 Hz that includes individual finger control. It never names the hand, gives no per-hand DoF count, and reports no success rate or trial count for any task. It is the maker describing its own unreleased hand, which is the evidence class the finding is about. The four research prototypes in Table V are in a different position, since the Faive Hand and LEAP Hand v2 Advanced account for three papers between them [47], [44].

Table V's emptiness is measurable and part of the same finding. Over the specification columns it prints, from DoF to price, 58 percent of its cells are values no source stated, against 42 percent for the hands that can be bought, and both captions state their own count. The hands with the highest advertised DoF counts have the least specification behind them.

The one hand that crosses the gap crosses it because its maker published rather than announced. ByteDexter reaches a paper through a ByteDance technical report stating 21 DoF per hand, 16 piezoresistive fingertip channels in the action vector and real-robot success rates [83], not through a product page. That mechanism is available to every vendor in Table V and none has used it.

The gap runs the other way too, and the last column of Table IV shows it. Eight hands that can be bought today or built from published designs take no papers at all: the Unitree Dex5 [84], the Tesollo DG-5F [85], the Proception ProHand [86], ORCA, RUKA, Ruka-v2, BiDexHand and DexHand. Cheapness is established for only three of those eight, because the price cell is empty for the rest. A reader choosing a hand on this corpus's evidence has four options with published precedent behind them, and only one of the four, LEAP, has both a published design and a stated price. Below those four the record thins fast, and 19 of the hands in Tables IV and V sit there on their specifications alone, with no paper in this corpus running on any of them.

Nineteen of the 33 hands in Table IV and Table V appear in no paper, and 11 of those are neither sold nor open and appear in none for that reason, so 33 is not the denominator for a software-lag claim: the eight hands of the paragraph above are. The rule that decides used from unused is one regular expression per hand against the papers' own hand field, in `tools/hand_usage.py`, so the partition can be recomputed rather than argued about. The lag runs through the tooling as well as the papers. Seven of the fifteen simulators ship a first-party hand model at all, six of those hands are real, Brax's being a synthetic claw, and every one of the six is an Allegro or a Shadow; only four of the 33 tabulated hands appear in any engine. Bench2Dex [87] meanwhile compares 12 hands and DexVerse [88] six without stating a DoF count for any. A useful next step is a conformance suite for hand models: one URDF or MJCF per hand, with fixed joint-limit, mass and collision checks and a published pass or fail per engine.

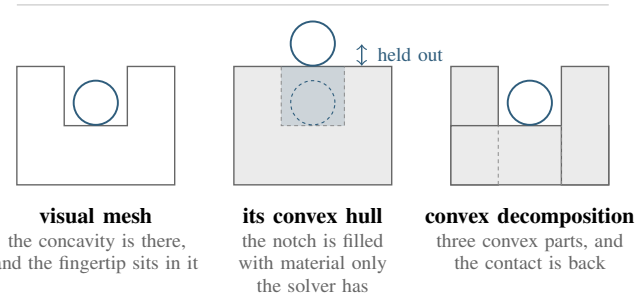


Fig. 4: The geometry the solver uses against the geometry the renderer draws. A single convex hull fills the concavity with material that exists only in the collision model, and holds the fingertip out of a contact the visual mesh allows.

IV. Simulators and the physics underneath

A. Requirements of a dexterous simulation

The clearest demonstration that hands are the hard case came from a benchmark that was not about hands. Erez et al. built a 35-DOF arm modeled on the Shadow Hand, closed it around a capsule with fixed spring-dampers, and asked four of the five engines for the largest timestep at which the object was still in the hand. The answers spanned a factor of 500, from milliseconds down to tens of microseconds, with MuJoCo at the slow-step end [89, Sec. IV-D]. Three caveats travel with that spread. The engines ran a deliberately restricted common model, hinge joints with sphere and capsule geometry and no meshes; the authors call it "an open question whether this model system can be tuned to work better in the gaming engines"; and they wrote MuJoCo and say so.

A grasp is hard for nameable reasons. It is many contacts at once, all persistent, all near stiction, on an object much lighter than the mechanism holding it. Persistence and multiplicity make the problem hyperstatic, and Le Lidec et al. show that per-contact solvers of the projected Gauss-Seidel family, and RaiSim's, then inject spurious jamming forces at stiction that vanish only once the object slides. The mass ratio makes it ill-conditioned, and in their stacked-cube test at a 10^{3} to 10^{-3} kg ratio those same methods fail to converge. Global methods with proximal regularisation stay robust in both cases [34, Sec. IV-A]. What degrades a truncated solve is therefore conditioning and redundancy, not stiffness, and a grasp supplies both by construction. Locomotion is forgiving by comparison. On flat ground the contact model and solver choice "hardly affects" their quadruped's tracked base velocity, and only on rough, slippery terrain do RaiSim and CCP deviate. A walking robot makes and breaks a few contacts against ground far heavier than itself. A hand does neither.

Speed enters by the same door. MuJoCo Playground reports that contact time scales with the number of possible contacts rather than the active ones, because JAX requires static shapes, which is why its tasks override the bound on contact points and the bound on geometry pairs by hand [90, Sec. VI]. Every override printed in that paper is a locomotion port. A hand's bound would have to be set the same way, for the worst case, but no hand configuration is printed.

Fig. 5 sets out the stages of one simulation step. Three of them make overlap, and they do not answer to the same knob. In an engine that enforces non-penetration at the velocity level, integration turns any residual approach velocity into overlap of order v times Δt; an engine that enforces the gap at the next configuration carries no such term, which is why Dojo's hard-contact NCP keeps its feet above the floor at every timestep it was tested at. Constraint assembly fixes the compliance a loaded contact then rests at. A truncated solver leaves a residual that grows with conditioning. Which of the three dominates in a grasp is not measured anywhere in this corpus, and the three are not ordered here. A fourth thing is not on the figure, because it is not a stage of the step and not a source of overlap: a mismatch between the geometry the solver uses and the geometry the renderer draws, which runs in both directions and is drawn in Fig. 4.

B. Contact models and solvers

Table VI, set in Appendix B, is the engine-by-engine comparison, one line per simulator, built only from what a note confirmed. Le Lidec et al. supply the taxonomy that organises it, checking each formulation against the Signorini condition, Coulomb's law, and the maximum dissipation principle. Linear complementarity, the family of Bullet, ODE, PhysX and DART, satisfies Signorini alone, because linearising the friction cone to a pyramid biases friction toward its corners. The cone complementarity problem satisfies the other two but relaxes Signorini, so contact acts at a distance of size $\Delta t \cdot \mu \cdot \|c_T\|$. The full nonlinear problem satisfies all three and is non-convex, by Table II of [34]. A contact model is not the algorithm that solves it. PhysX linearises the cone, which places its model in the LCP family, and it does not solve an LCP. Its Temporal Gauss-Seidel scheme folds substepping into the Gauss-Seidel sweep and exposes position and velocity iteration counts separately [91, Sec. 3].

MuJoCo is inside that taxonomy. Le Lidec et al. classify it as CCP-MuJoCo, a convex relaxation solved by a Newton method on the primal QCQP, in the same family as CCP-Drake, the SAP-style scheme, by Table III of [34]. The relaxation makes two errors of opposite sign in one engine. A loaded contact carries a violation, and a sliding contact carries force at a positive gap. Drake's SAP inherits the same pair, and its gliding effect at distance $\varphi \approx \delta t \cdot \mu \cdot \|v_t\|$ "unfortunately does not go away as $\delta t \to 0$" [92, Sec. V-A]. Le Lidec et al. call that compliance a "numerical trick designed to circumvent the issues due to hyper-staticity or ill-conditioning at the cost of impairing the simulation".

The depth a loaded contact carries is a chosen number, not a property of the engine. In a regularised formulation the steady-state violation at a contact is the normal load times the compliance: zero at zero load, growing with what the contact carries. MuJoCo drives that violation with a critically damped stabiliser whose steady-state depth, for an object resting under gravity, has a closed form independent of the object's mass [93, Sec. V]. Whoever sets the stiffness sets the depth.

Published penetration figures do not compare with each other, and the reason is worth knowing before anyone quotes one. Dojo solves a hard-contact problem and stays above the floor on an Atlas drop test where MuJoCo overlaps by tens of millimetres [94, Sec. V-A]; Drake's figures for the same quantity are orders of magnitude smaller, and they are analytical bounds for a point mass at rest rather than measured depths [92, Sec. V-B]; ComFree-Sim reports a depth directly, on small convex primitives dropped under their own weight [95, Sec. IV-A]. A point-mass bound, a humanoid drop transient and a dropped primitive differ in load, inertia and regime, so the distances between those figures measure nothing about each other. What the set does establish is the part that matters for a hand: in a compliant formulation the depth follows from a stiffness somebody chose, and the GPU era moved that knob rather than removing it. The oldest engine comparison here puts the consequence plainly, reporting that engines pushed past their stable step "effectively simulate a different physics model which can no longer hold the object" [89, Sec. IV-D]. No paper in this corpus reports the depth for a fingertip loaded by a grasp, which is the one configuration a dexterous result depends on.

Table VI's most interesting column is the one that records whether an engine's own paper reports a penetration depth, and it is nearly empty. Two engines report one, and both are engines whose paper is about contact accuracy. The word "penetration" appears nowhere in the Isaac Gym paper, and Isaac Lab's contact sensor reports force, duration and an average contact point with no measure of contact quality at all. That is a documentation gap rather than a capability gap: NVIDIA's own benchmark repository already computes an interpenetration depth per environment for its assembly tasks and scales the reward down as the depth approaches a millimetre. The quantity is a few lines of code away in the engine most of this literature trains in.

The rest of Table VI is largely empty, and the emptiness is a result: 38 percent of the cells in its property columns hold a value no source stated, which is what its caption counts. No engine paper states a default physics timestep, and the three that report one report it for a named experiment, so the timestep column carries experiment settings rather than defaults. Fewer than half the engines ship any dexterous hand. Two columns only the repository tabulation carries are worth the trip: the solver iteration count, which four engines state, and the license, which is the question a reader choosing an engine actually has and which almost none of them answers.

C. GPU-parallel engines and their throughput

Isaac Gym set the pattern. Physics, observations, rewards and actions stay on the GPU, and PhysX resolves contacts with the Temporal Gauss-Seidel sweep described above. Its per-task timesteps are published, which is rare: the Shadow Hand runs a 1/120 s physics step under a 1/60 s control step, or 1/20 s in the OpenAI variant. The result that reorganised the field is that reproducing OpenAI's Shadow Hand cube reorientation took under an hour on one A100, against 30 hours on 6144 CPU cores and 8 V100s [91, Sec. 6.4.1]. Thirty-five of the papers examined run on it.

Orbit and Isaac Lab moved the stack to PhysX 5, and the dexterous offering is thinner than the predecessor's: the first-

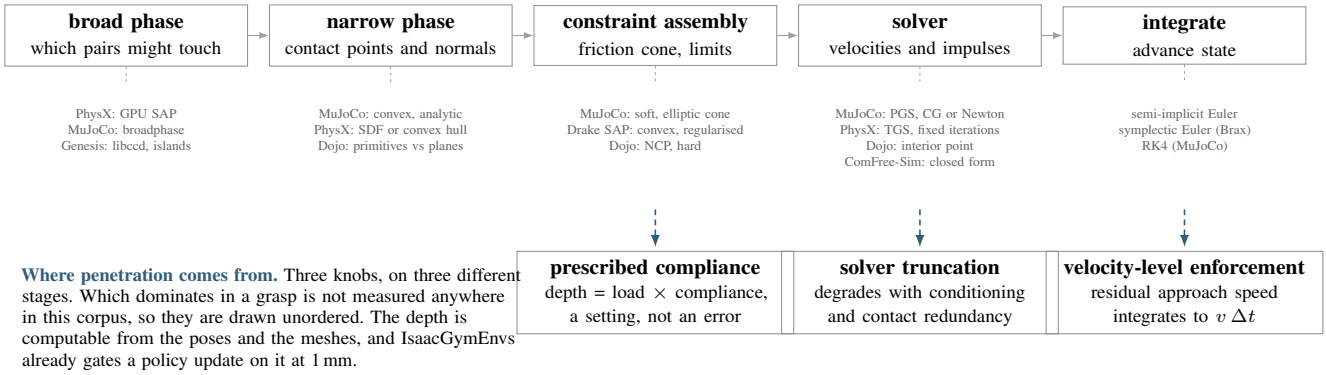


Fig. 5: One simulation step, and where engines differ: the stages every rigid-body engine runs, with the choices that separate the engines used for hands under each; dashed arrows mark the three stages that overlap.

party suite is lifting, grasping and reorienting with the KUKA Allegro hand, while Shadow and Ability hands appear only through third-party grasp evaluation [96, Sec. 7.2.3]. MJX and MuJoCo Playground took the other route, putting MuJoCo's own solver on the GPU through JAX at the cost described above. MuJoCo Warp ports the same solref and solimp soft-constraint entries to NVIDIA Warp, and its README states that the legacy PGS solver and the noslip pass are unsupported and that differentiability "is not yet available" [97]. GPU MuJoCo is not a differentiable MuJoCo. Genesis is the broadest, pairing a rigid constraint solver with FEM, MPM, PBD and SPH solvers and a differentiable backward path, and it ships a Shadow Hand. Its README quotes no throughput number at all, so the large FPS figures that circulate for Genesis trace to nothing in [98].

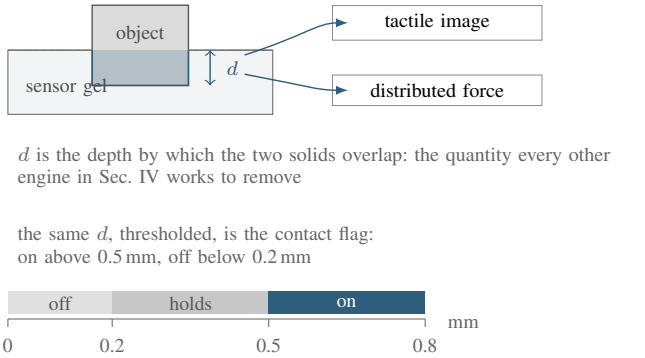
Newton changes how these names should be read. It is explicitly multi-solver: SolverMuJoCo wraps MuJoCo Warp, SolverKamino is a proximal-ADMM and dual-variational-inequality contact solver, and XPBD and VBD do position-based constraint projection [99]. Its contact model is solver-dependent, not a property of the engine, which is how Table VI records it. Isaac Lab already offers a Newton–MuJoCo Warp backend in its hands demo, and its roadmap announces Newton integration. An experiment reported as Isaac Lab may be running PhysX 5 with TGS, or MuJoCo's soft-constraint entries under Warp, and those two make different contact errors. The name also fails to fix the physics inside one engine. MuJoCo's convex solver is a family, interior point or projected Newton, conjugate gradient or Gauss-Seidel [100, Sec. II-D], and MuJoCo Warp supports neither PGS nor the noslip pass. PhysX 4 and PhysX 5 differ in whether non-convex rigid bodies get SDF collision, which is a contact-geometry difference between Isaac Gym and Isaac Lab under one vendor name [101], [96]. Naming a simulator no longer names its physics. Papers should report the backend and the solver beside the framework.

Four headline figures measure four different quantities. Isaac Gym reports parallel environment steps per second with physics, observations and rewards on device: 150,000 for the Shadow Hand at 16,384 environments on one A100. Isaac Lab reports frames per second in training, which includes the learning update: over 900,000 for the DextrAH teacher task at 16,384 environments on eight RTX Pro 6000 GPUs, with no single-GPU dexterous number anywhere in the paper. ManiSkill3 reports "up to 30,000+ FPS" with RGBD and segmentation on one RTX 4090, measured over 1000 random actions with reward and termination removed from the timing, and does not state the environment count behind the figure [102]. Orbit reports a 125,000 FPS physics-only ceiling on an RTX 3090, also with no environment count [101, Sec. VII].

The one number reported cleanly enough to reuse is MuJoCo Playground's, in training steps per second on one GPU over five seeds. A hand reorientation task runs about six times slower there than a parallel-jaw pushing task on the same hardware in the same codebase. That factor is not a property of the task alone: Playground tunes solver iterations, line-search iterations, timestep and contact bounds per environment, in its own Table III [90], and does not print the two configurations side by side, which is what makes even those two figures incomparable. A training-loop rate is also mostly not a measurement of physics; on Playground's own breakdown the policy update dominates the frame and physics is a few percent of it.

Cross-framework comparisons add further free parameters. One widely cited table takes the best result over a range of environment counts for each system [103, Table 1a], which tunes a free parameter per competitor, and the two systems being compared are not the same kind of thing: one runs rigid-body physics on CPU workers behind a shared render server and the other on the GPU. Because the measured loop includes rendering at a fixed control rate, what is compared is a visual sample-collection loop rather than two physics engines. Its authors say so themselves, and Playground is equally explicit that its own cross-simulator plot borrows two of its numbers from another paper. Several sources give no figure at all, pointing instead at an external dashboard or omitting throughput from their documentation entirely.

The failure reaches past the engine papers into the methods. Of the 47 corpus method papers that name a GPU-batched simulator, 19 state no environment count anywhere, and Isaac Lab's own paper states no physics timestep, no decimation and no solver iteration count for any task. A throughput figure is interpretable only with the environment count, the GPU, the timesteps, the solver iteration budget, and a statement of whether rendering and the learning update sit inside the



The inversion. Everywhere else in this survey the overlap is the error. For a compliant tactile sensor it is the signal, so the quantity a policy elsewhere is penalised for producing is the one this sensor is calibrated to read.

Fig. 6: A compliant tactile sensor reads the overlap depth. The tactile image, the distributed force and the binary contact flag are all functions of the one quantity every other engine in this section works to drive to zero.

measurement. Almost nobody reports all five.

D. Tactile simulation and the sim-to-real gap

The three tactile simulators in this corpus calibrate against three different things, and none of them is a manipulation outcome. TACTO is a rendering layer over a host engine, by default PyBullet's rigid contact model. It reads post-solve link poses and the engine's reported normal force and maps that force to gel-mesh deformation at the rendering level, so it contributes no contact physics of its own. Its only sim-to-real number is a tactile pose-estimation task, at 1.66 ± 0.16 mm with color-jitter augmentation against 0.76 ± 0.07 mm for a model trained on 128 real datapoints, by Table II of [104].

Taxim is example-based rather than simulated, calibrated from real indentations, and it beats TACTO on every optical-similarity metric against real images. Its marker-motion model is calibrated against a finite-element reference, and it tracks that reference about three times more tightly than the reference tracks reality, so its residual is bounded below by its own calibration target. The authors also state that they simulate quasi-static contact only, with slip left as future work [105, Sec. V]. Slip is what a hand needs.

Tactile Genesis inverts the thread. It implements two penetration-depth backends on the collision geometry, an analytic SDF query and a BVH raycast, and uses the depth as the sensor signal rather than as an error, so the quantity every other engine treats as a defect sets this one's operating point.

Most of what is written about this gap is attribution without measurement. PenSpin asserts that the pure physics gap "cannot be bridged by extensive domain randomization alone" while reporting no experiment that isolates it [72]. MuJoCo Playground attributes its LEAP hand failures to physical flex in low-cost hardware and names more accurate collision geometries as the fix, without measuring either. DeXtreme lists four candidate causes for its shortfall, including a malfunctioning Allegro thumb used in most trials, and disambiguates none by experiment [14].

What has been measured is narrower, and Section VII collects the transfer numbers. The one result that links a quantified model error to transfer ranks system-identification runs by dynamics-model error and pairs each with real success: the best-identified model grasps 8 of 10 attempts, the worst 0 of 10 [106, Table 1]. Where a cause has been pinned down elsewhere it is usually perception or actuation rather than contact, with pose estimators and camera trackers named as the unmodelled source in the real numbers [107, App. C].

The one paper that looks at the contact side is usually read backwards. DeXtreme's real-to-sim replay interpenetrated because the replayed poses carried the pose estimator's error, not because the contact model failed, and the paper says so: it could not calibrate physics against the real system because the state it replayed was already wrong. The commonly cited reading, that the contact model was at fault, is not what the sentence says.

V. Training a dexterous policy

A. Sources of supervision

A dexterous policy is defined by what supervises it, and four sources are in use across the corpus's 113 papers: a reward function supervises reinforcement learning, a human demonstration supervises imitation, a human reference trajectory supervises a physics-based tracker, which is imitation with a simulator in the loop, and nothing supervises a model-based planner, which is handed a cost and a model instead.

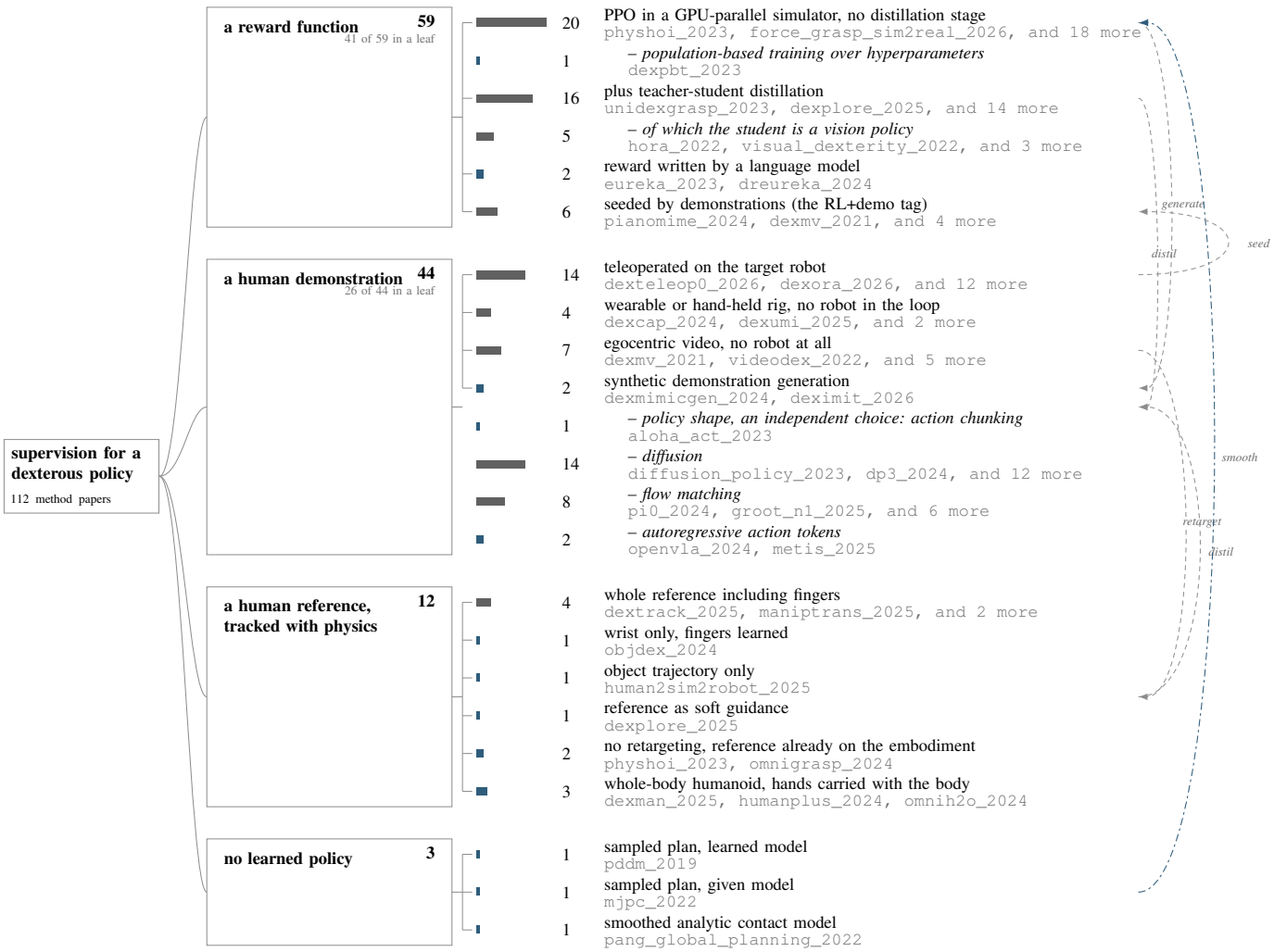
The labels do not partition the corpus. Reinforcement learning is the largest at 54 of the 113 papers, behavior cloning next at 27, distillation at 23, and ten further tags cover the rest. They sum to far more than 113, because most methods published since 2024 sit on two branches at once: a reinforcement-learning teacher and a behavior-cloned student are one paper, not two. Fig. 7 draws the tree and the cross-links. The repository carries the paper-by-paper version of the same thing, one line per method, and that is what to scan when looking for work comparable to your own.

What the deployed policy looks like once training is done matters more than the name a paper gives itself, and on that axis the field has converged hard. Section V-E shows why.

The emptiest columns of the method tabulation are the ones a reader most needs. Only 70 of the 113 papers state how many real trials are behind their headline number, and fewer still state an environment count or how many unseen objects were tested; Section VII-A takes up what that means for comparison. A mostly empty line is not a weak method, but it is one that cannot be compared with any other here. All 113 papers carry the same columns, and Appendix A says where they are and how to read them.

B. Reinforcement learning

Sixty of the 113 papers learn from a reward. Forty-nine name PPO as the algorithm, and the next most frequent, DAPG, appears four times. Forty-three run their own experiments in a GPU-parallel simulator from the Isaac family or Genesis, and 36 do both. Only 32 papers state an environment count,



Eight further rows carry a trajectory-optimisation or MPC tag inside a learned pipeline and are counted on the branch that learns. The double-dashed edge is the one overlap that is a theorem rather than a pipeline: `pang_global_planning_2022` proves that a policy gradient and an analytic log-barrier relaxation compute the same local model of contact.

Fig. 7: What supervises a dexterous policy, subdivided to where the 113 method papers actually differ; each leaf names two of its corpus keys and counts the rest, bars scaled by paper count, dashed curves marking overlap.

with a median of 8192 and a maximum of 64000. Twenty-three papers distill a privileged teacher into a deployable student, and 16 combine all three of PPO, a GPU simulator and distillation. That 16-paper intersection is the recipe as it is actually practised.

The hand is almost always an Allegro. It appears in 35 of them, and its hold on in-hand reorientation is tighter still: more than half of those papers use it, against roughly a fifth for the Shadow and an eighth for LEAP.

Privilege is handled two ways. OpenAI [2] gives the value network object and target orientation, joint angles, joint velocities and object velocities the policy never sees, with a footnote recording that current object orientation was left out of the policy inputs by accident, and DeXtreme [14] uses the same asymmetric critic with a 2048-unit LSTM against the actor's 1024. Hora [75] instead compresses nine privileged object properties into an eight-dimensional vector regressed from 30 steps of proprioception, and Visual Dexterity [16] distills twice, from a state teacher to a synthetic point cloud and then to a rendered one, for a fivefold speedup.

Domain randomisation is the part of the recipe with the least discipline. The best practice in the corpus makes it adaptive, widening each range when performance at its boundary is good enough and narrowing it when it is not, and publishing the ranges that search discovered [108], [14]. Below that standard, two shipped configurations disagree with their own papers about what was randomised at all: one ships a joint-noise range four times wider than the paper states, in a revision its own README says does not reproduce the paper [75], and the other ships a disturbance force of zero where the paper describes one [72]. Both are in Table II with the file and the two values.

The repository puts the 21 in-hand reorientation methods against nine recurring term families and marks each cell by

where the term was found: in the paper, in the code, in both, or in the code with every shipped configuration setting its weight to zero.

The families are not equally popular. Almost every method has a goal or rotation tracking term, which is the task itself. After that the common families are an effort penalty, a sparse success bonus, an object-velocity penalty and a drop penalty, each in about half of them. Only four carry a contact or force term at all.

Three marks an earlier draft recorded as *code* now read *code (0)*, because in each the term is in the released code with every shipped configuration setting its weight to zero: DeXtreme [14]'s timeout reward, DexPBT [109]'s fall penalty, and PenSpin [72]'s action penalty. Marking them *code* would tell a reader the code optimises something the paper does not state, and leaving them unmarked would hide a term that is in the file. The repository carries the config key and the zero beside each of the three marks.

Only three of those four contact terms are what they look like. AnyRotate [39] scores good and bad fingertip contacts, Lin et al. [110] reward a friction-cone wrench margin, and Zhao et al. [111] track a commanded grasp force. The fourth [16] penalises the *object* touching the table, which is task shaping against using the table as a third finger rather than a statement about hand–object contact, and the flag that enables it defaults to off. So three of the 21 reorientation methods put a hand–object contact quantity in the reward, and none puts interpenetration in it: the one method that penalises interpenetration does so while building its references offline, not in the policy's reward [112]. The physical quality of the contact is not something this literature optimises.

Term counts range from one to ten. DemoStart [113] gives a single binary terminal reward of 1 and lets a curriculum over demonstration states do the shaping. AnyRotate [39] and ViserDex [114] each publish ten weighted terms. ViserDex is the cleanest specification of the 21, with every term, weight and equation in one appendix table, and a statement that the same weights are used for every object.

Weights are frequently unrecoverable. Three of the 21 papers give no numeric weight that survives PDF conversion, because the equation blocks are images, which is a limit of this survey and is recorded as such. Robot Synesthesia [70] is a worse case. Its six coefficients are described only as "tuned hyperparameters" and no number is printed anywhere in the paper, so the reward it reports cannot be reconstructed from the paper. DexReMoE [115] names an angular-velocity penalty weight in prose with no entry in its hyperparameter table, and DexTrack [116] names an affinity reward that is missing from its weight table.

Curricula in this literature relax physics or tighten tolerances. DexPBT [109] tightens a success tolerance from 0.075 to 0.01 by a factor of 0.9 every 3000 environment steps once three successes are logged, and ManipTrans [17] starts at zero gravity and high friction and restores both while narrowing a fingertip threshold from 6 cm to 4 cm. DexMachina [18] attaches virtual spring-damper controllers to the object, lets them do the task at first, then decays their gains to zero. The two that ablate the curriculum report it is load-bearing: RotateIt [71] sets its rotation-penalty weight to zero and ramps it to -0.1, because applying it from the start makes the policy learn only to hold the object still, and ViserDex [114], whose performance-based curriculum runs over regularisation weight, action latency and time between successes, fails completely without the regularisation component.

Population-based training appears once. DexPBT [109] trains populations of up to 32 agents, mutating the middle third of each population and replacing the bottom with mutated copies of the top, and reports 0.32 trillion environment steps for its largest run. The technique is routine outside robotics and nothing else in this corpus uses it.

A language model can write the reward outright, which removes the hand-tuning and replaces it with a search whose objective nobody has read. Eureka [15] has GPT-4 write the reward function directly, constrained to return a total and a dictionary of named components, and feeds per-component statistics back between iterations. Its appendix gives an example whose reward correlates at -0.26 with the human-written one and still scores 1.45 on the human-normalized metric, the strongest argument in the corpus that hand-tuned weights are not a ceiling. DrEureka [117] moves the safety requirement into the prompt instead of a penalty term, asking in words for a cube rotating at about 0.25 radians per second with fingers penalized for leaving their initial pose. Neither can be checked against a file. Eureka's generated rewards are written at run time to a gitignored directory and never committed, so the reward behind any reported number cannot be recovered from the repository, though its README documents the paper's budget of five iterations and sixteen samples as the default and marks the shipped one-iteration block a fast-test preset. DrEureka's repository contains only the locomotion and globe-walking trees, so its LEAP-hand reward has no code at all.

Reinforcement learning stalls on objectives that are not learnable as stated. AnyRotate [39] found the angular-velocity objective unlearnable in the multi-axis setting and replaced it with a moving keypoint target. RotateIt [71] reports that its multi-axis policy "does not converge when training with only reinforcement learning" and adds an imitation loss against single-axis oracles. Eureka [15] cannot spin a pen from scratch and needs a pretrain-then-finetune split, with the from-scratch ablation failing to complete one cycle. Section V-C's contact-graph term exists because of the general shape of the problem: contact moves the object off the reference, so return goes down, so the policy learns not to touch the object.

It stalls on geometry too. Hora [75] fails on objects under 4 cm across because the fingers collide with each other. What it does not stall on is publication: the real-trial counts behind these headlines are small enough that Section VII-A treats them as the reporting problem they are.

C. Learning from human data

Thirty papers in this corpus describe a system for getting human motion onto a robot hand; the repository lists them with the retargeting objective compressed to one clause each.

Retargeting splits four ways. The dominant family matches task-space vectors between the human and the robot hand,

weighting fingertip distances and adding a minimum separation between the primary fingers, solved online; DexPilot [118] set that pattern and four later systems use the same formulation [119], [120], [43], [121]. The second copies joint angles wherever the kinematics allow it and solves only the thumb by inverse kinematics [122]. The third learns the map offline, which buys about an order of magnitude in rate over online optimisation and raises configuration-space coverage from 38 to 90 percent [123]. The fourth removes retargeting from the loop altogether, building an exoskeleton whose fingertip workspace is designed to match the target hand and regressing encoder values straight to motor values [69].

What that tabulation does not carry is the point. Fewer than half of the teleoperation papers state a latency at all, and only a handful state a rig cost. Where costs are stated they are low and falling, from a few hundred dollars for a headset or a glove [122], [124] to a few thousand for a full mocap backpack [125], [44]. Latency is not one quantity either: some figures are end to end [118], [122], others are a single retargeting stage [123], [119], and several cells hold a control rate instead, so the column's spread is not a range. Fewer than half report how many trajectories they collected, and fewer still report hours of data.

Where throughput is measured, human collection beats teleoperation by a wide margin. Qiu et al. [126] time a grasp demonstration at 3.79 seconds for a bare human and 19.72 seconds through a teleoperated humanoid, and a pour at 4.81 against 37.31 seconds. It attributes the gap to retargeting latency and to the workspace of a 7-DoF arm rather than to anything about the hand.

Four architectures cover the corpus. Action chunking came from [127], which predicts a chunk of joint targets with a CVAE and combines overlapping chunks by exponentially weighted ensembling, reaching 80 to 90 percent on fine bimanual tasks from about 50 demonstrations each. Diffusion came from Diffusion Policy [128], which denoises an action sequence and reports a 46.9 percent average success improvement over LSTM-GMM, IBC and BET across 15 tasks. 3D Diffusion Policy [129] swaps the image encoder for a small point-cloud encoder and reports 74.4 percent against 59.8 percent across 72 simulated tasks, and 85.0 against 35.0 percent on four real tasks from 40 demonstrations each. Flow matching is the 2024-onward default for large models and carries eight papers. Discrete action tokens are the fourth, used by [24] and [130], and are the only one of the four that needs no continuous head at all.

Data generation is a separate lever and is underrated. DexMimicGen [45] turns 60 human demonstrations into 21,000 simulated ones, and a real two-handed humanoid trained on the generated set succeeds where the same policy trained on the source demonstrations never does. Dex1B [131] iterates optimisation, a proposal model and a simulator filter into roughly a billion synthetic grasps, and beats the prior best by a wide margin on its own benchmark.

Seven papers train a dexterous-hand policy from video with no teleoperation at any stage. DexMV [132] retargets 700 self-recorded demonstrations by matching palm-to-fingertip task-space vectors and estimating actions by inverse dynamics, VideoDex [133] mines 965 trajectories from EpicKitchens to pretrain a neural dynamic policy, and DexVIP [134] goes further into the reward, clustering 715 curated HowTo100M frames into one consensus grasp pose per object category and adding it as a reward term. OKAMI [135], Lum et al. [136] and Guzey et al. [137] work from a single video each, and Goswami et al. [138] pretrain a world model on 829 hours of EgoDex.

Three papers measure the gap directly rather than asserting it. OKAMI [135] runs the same reference-plan and warping pipeline with and without human body and hand retargeting, and the version without it loses 75.0, 42.1, 82.0 and 74.0 points across four task and setting pairs. That is the largest clean embodiment-gap number in the corpus.

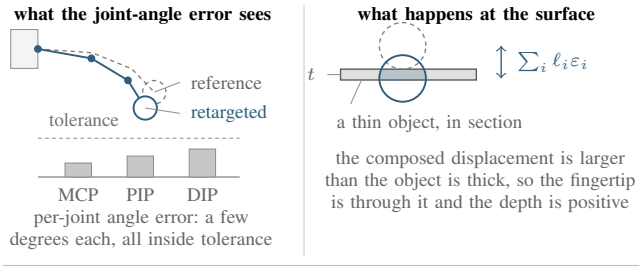
Qiu et al. [126] ablate two mitigations separately on the same vertical-grasp task. Removing the interpolation that slows human trajectories down cuts success to a quarter of the full method's, and removing the unified state space collapses it to zero, because the policy is handed a shortcut for telling the two embodiments apart. EgoZero [139] reports the sharpest version of the same effect: removing its 3D augmentation drops every one of its seven tasks to no successes at all, and substituting monocular depth for triangulated depth does the same, because the best metric depth models it tested carry more than 5 cm of error.

Chen et al. [140] supply the counterexample. Their method retargets only the wrist and lets reinforcement learning discover finger motion, and their ablation that adds a fingertip-matching reward "does not yield benefits and even leads to lower performance". Richer human correspondence is not uniformly better, and this is the one result in the corpus that says so with an ablation.

Twelve papers sit in the human-reference tracking family. Eight track a human hand on an object and are covered here; the other four sit at the edges, Lum et al. [136] tracking only the object's trajectory, DexMan [141] retargeting bimanual video onto a full humanoid, and HumanPlus [26] and OmniH2O [27] tracking whole-body motion. A reference is retargeted onto the robot and a policy trained to make the simulated body follow it: the reference supplies the shaping reward engineering would otherwise have to invent, and the simulator the physical consistency pure imitation lacks.

The multiplicative reward that most human-reference methods now use has a single source. PhysHOI [142] is the origin of the form. It multiplies a body term, an object term, an interaction-graph term and a contact-graph term, and reaches 95.4 percent success on GRAB against 27.0 percent for a DeepMimic baseline. The contact-graph term exists to stop the policy learning not to touch the object. Omnigrasp [11] replaces the action space with a pretrained latent motion prior and reports 94.6 percent grasp success and 84.8 percent trajectory success on GRAB, while stating outright that it omits penetration metrics.

The next step was to stop treating demonstrations as fixed and let the method generate its own. DexTrack [116] adds an imitation loss to PPO and mines its own demonstrations through a homotopy search over easier neighbouring trajectories, beating the best baseline by about eight points at either



The gap. Joint-angle error is a sum over joints and carries no term for the object, so it cannot see the right-hand panel at all. Errors small enough to pass at every joint compose along the finger into a fingertip displacement larger than a thin object is thick, and the pose is scored as good with the object inside the finger. Sec. V's one method that measures penetration measures it on the reference, before the policy runs.

Fig. 8: A retargeted pose can be scored good and still be inside the object. The joint-angle error is a sum over joints with no term for the object; a few degrees at each joint compose along the finger into a fingertip displacement larger than a thin object is thick.

threshold. It has a penetration-depth formula and applies it only to its input references, never to its own rollouts, and presents tolerance of "severe hand-object penetrations" as robustness. ManipTrans [17] freezes a generalist hand imitator and trains a per-task residual on top, with single-hand success about half again its bimanual figure. Its stance on contact is to raise the friction coefficient above the real value rather than model skin deformation, and its real deployment is open-loop replay.

Two papers in this corpus ran the same curriculum and disagreed about whether it helps. DexMachina [18] moves to Genesis with 12,000 environments and six hand URDFs, and drives the object with virtual controllers that decay to zero. Its re-implementation of [17]'s curriculum does not beat no curriculum on its own setup, which is a direct disagreement between two papers worth quoting in both directions. Dexplore [74] drops the explicit retargeting stage and treats the reference as soft guidance, with termination thresholds derived from the failure rate, reaching 87.7 percent on GRAB with an Inspire hand.

TopoRetarget [143] is the only corpus method that quantifies penetration as an evaluation quantity, reporting maximum depth and the fraction of frames past 2 mm, and constraining it during retargeting with a 1 mm soft tolerance and a 30 mm hard bound. It reaches 87.5 percent on its own pen-spinning set against 46.9 for the best baseline. It does not re-measure penetration after the tracking policy runs, so the property it constrains is a property of the reference and not of the behavior. Fig. 8 draws why the usual retargeting objective cannot see the failure at all: joint-angle error is summed over joints and carries no term for the object, so errors small enough to pass at every joint compose along the finger into a fingertip displacement larger than a thin object is thick, and the pose is scored as good with the object inside the finger.

The pattern holds across all eight, and it is the reference-versus-rollout split in its sharpest form. Penetration is handled at the reference, if at all, and never at the rollout. Chen et al. [140] complete the set with the only real-robot numbers among them, from 100 percent on a microwave and a laptop down to 41.2 percent on a ketchup bottle over 20 trials each.

Human data does not port across hands, and the map is usually left unstated. Fifty-four papers use human data and name 45 distinct hand strings between them, and twenty of those appear among the 30 teleoperation systems with a stated retargeting objective. The other 33 never say how the human motion reached the hand, and the stated objectives do not converge, from a fingertip keypoint-vector energy in [119] to a per-joint regression from exoskeleton encoders in [69] to no retargeting at all in [139]. A shared benchmark could make these mappings comparable. TopoRetarget's format [143] is a useful starting point: it reports contact precision, contact alignment, maximum penetration and the share of frames past 2 mm, per hand and per dataset.

D. Generalist and vision-language-action policies

The finding is the size of the hand. Eighteen papers carry the generalist tag. Fourteen of them settle whether the reported evaluation ran on a multi-fingered hand. Eight of those fourteen did, six report no hand result at all, and the remaining four never say, GR00T N1.6 [28] naming no end effector anywhere on its page.

Five of the eight state a hand size, and that comparison has to be made in actuated degrees of freedom rather than joints, for the reason Section III opens with. Four of the five use a six-actuator hand and one a twelve, so their median is six. The reinforcement-learning literature of Section V-B works an order of magnitude above that: its median hand actuates 16 joints. No generalist policy here that settles the question evaluates on a hand in that band, and the two that would reach it on paper are the two that never say whether the hand was in the evaluation at all.

The six with no hand result are parallel-jaw throughout (π_0 [21], whose released code encodes each ALOHA gripper as one scalar, $\pi_{0.5}$ [22], $\pi^*_{0.6}$ [23], OpenVLA [24], Helix [82], which claims individual-finger control of a 35-DoF upper body with no success rate or trial count for any task, and Gemini Robotics [144], whose Apollo hand appears in a figure caption with no number attached). Its successor [25] closed that gap with Apollo progress scores, not success rates, of 0.74 down to 0.62 across five generalisation axes. DexVLA [48] is the borderline case the other way: a hand in one real-robot task family, grippers in the other five and in all 91 pretraining tasks.

The released code also drifts. GR00T N1 [145]'s repository is a later N1.7 generation with a different backbone, a different action horizon and embodiments the paper does not describe, and the $\pi_{0.5}$ repository [22] supports only the flow-matching head, not the hybrid recipe the paper describes.

The gap is one of size, not absence. The generalist policies that do touch a hand run it at roughly a third of the actuation the reinforcement-learning literature assumes, and the papers that never settle the question are the smaller half of the same gap. The standard generalist suite, with the hand's actuated degrees of freedom and its vendor printed beside the number, should include at least one dexterous-hand task.

E. Model-based baselines and hybrids

Three corpus methods solve dexterous tasks without learning a policy, and they are the control group the rest of this section

lacks.

Not every method here learns a policy. A smaller line plans through a model instead, and reaches dexterous tasks on a fraction of the data. PDDM [107] learns an ensemble of dynamics models and plans through it with an MPPI-style optimiser, replanning every step, on a few hours of data in simulation and a few more on a real Shadow Hand. Predictive Sampling [146] removes the learned model too, sampling rollouts through MuJoCo itself at millisecond latency, and reorients a cube in real time from scratch. It also reports no success criterion, no trial count and no real-robot result.

The work of Pang et al. [35] is the most substantial of the three. It proves that the randomised smoothing implicit in reinforcement learning and an analytic log-barrier relaxation compute the same local linear model of contact, then uses the analytic version inside a trajectory optimiser and an RRT. Allegro in-hand rotation takes 19.59 seconds to optimize and its plate-pickup task 117.16 seconds of planning on a 16-core CPU, against the GPU-days of Section V-B, and it is the only method here that imposes non-penetration as a hard constraint. Its own limitation is the honest part: its 3D systems transfer to hardware far worse than its 2D ones, because the quasi-dynamic assumption breaks and planned grasps miss contacts under a second-order solver.

The recurring shape is reinforcement learning in simulation distilled into a policy that looks like an imitation policy, and it appears in four variants. The first distills a privileged teacher into a vision student inside one paper, which is [75], [16], [71], [70] and [114]. The second uses reinforcement learning as a demonstration factory. DexTrack [116] mines demonstrations with per-trajectory RL and trains a generalist on them, and ManipTrans [17] does the same to build a 3.3K-episode dataset. DexterityGen [147] goes further, training a diffusion controller on ten billion simulated grasp-to-grasp transitions and projecting a human teleoperator's coarse command onto it, taking four reorientation tasks from 0 out of 20 under raw teleoperation to 12, 13, 10 and 9 out of 20. The third generates data without reinforcement learning at all, which is [45], [131] and [148]; Dex1B's entry is classed as a dataset, so Fig. 7's leaf, which counts papers, holds the other two. The fourth wraps an RL policy in a residual, which is [149] at 88.8 percent over 3,200 objects in 12 GPU-hours.

In the variants that dominate 2025 and 2026 work, the reward is no longer the objective of the deployed policy. It is the objective of the process that produced the deployed policy's training data. Reward engineering has moved upstream into data curation, and every reward-shaping pathology in Section V-B now reaches the shipped policy through a dataset rather than a gradient.

F. Paper against released code

A paper that states its reward and also ships the code that trained it can be read twice, and the two readings sometimes disagree. In eight of the papers here they state different things. The sharpest case is PhysHOI [142]: its paper assigns the object's rotation error a nonzero weight on the GRAB dataset, and in the released reward function that error is set to zero

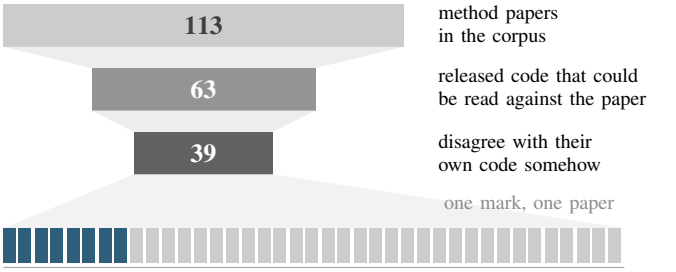


Fig. 9: Papers and their own released repositories state different things, and the funnel ends at the 8 the survey stands behind as contradictions. Table II names them, with the file and both values.

with the computation that would produce it commented out on the same line, so no weight on it could move the reward. Its reported success criterion uses position alone, so a run of that file would not expose the difference either. Table II states all eight the same way, each as a repository, a revision, a file, the value the paper prints and the value the file contains, so a reader with a browser can confirm or refute any line of it without this survey and without anyone's agreement.

What the comparison does not do is name a cause. A repository at some revision is not the code that produced a paper's numbers, and nothing here is a claim about intent or about work that released nothing to read.

The authors of every paper named in the table were written to on 19 September 2026, before this survey was posted, with the claim, its evidence and the sentences the survey would print, and were given until 10 October 2026 to answer. One reply changed a comparison and removed it from the count. One letter was redirected by a coauthor to the first author of that paper, who had not been written to and whom the window closed before anyone asked, which is this survey's miss rather than a non-reply. The rest are recorded as they came, and silence is recorded as silence rather than as agreement. The letters as they went out are in `outreach/`, so a reader can see exactly what every author was asked, and `outreach/RECIPIENTS.md` records who each one went to and how the address was found. A disputed line is corrected the same way it was made, in public: the route is the address in the author block, and Appendix C prints every comparison beside the review note that revised it.

The practical lesson for a reader of this literature is small and worth keeping: read the reward function before the reward table, and treat a printed weight as a hypothesis about the code. Future papers can remove the ambiguity altogether by generating the reward table from the configuration used for training and citing the revision it came from.

VI. Bimanual dexterous manipulation

Two hands are not two instances of the same problem. They share one object, they can collide with each other, and the action space doubles, so neither the hardware result nor the policy carries over from one hand. Counting the work is the first difficulty, because the widest count, the 53

TABLE II: The 8 rows where a paper and its released repository state different values, at the commit fetched, with confidence.

method	conf.	repository, fetched commit	file in it, and where	what the paper prints	what that file contains
PDDM [107]	high	`google-research/pddm` `06b88cdbaf`	`pddm/envs/cube/cube_env.py` `_get_obs`	observation dimension 46 for in-hand reorientation, Table 2	the six fields `_get_obs` concatenates sum to 39 on the code's own inline comments
DeXtreme [14]	high	`isaac-sim/IsaacGymEnvs` `aeed298638`	`isaacgymenvs/cfg/task/AllegroHandDextremeADR.yaml` `actionDeltaPenaltyScale`	action-delta penalty weight -0.25, Table 2	`actionDeltaPenaltyScale: -0.2`, and -0.01 in the ManualDR yaml beside it
Visual Dexterity [16]	high	`Improbable-AI/dexenv` `ad9634e9d2`	`dexenv/conf/dclaw.yaml` `alg.num_envs`	32,000 teacher training environments, Table S1	`num_envs: 8000`, and 16384 in the parent `hand_default.yaml`
DexPBT [109]	high	`NVIDIA-Omniverse/IsaacGymEnvs` `aeed298638`	`isaacgymenvs/tasks/allegro_kuka/allegro_kuka_base.py` `compute_kuka_reward`	four staged reward terms, Sec. III-C	eight summed components, `hand_delta_penalty` multiplied by 0 with the comment `currently disabled`
PhysHOI [142]	high	`wyhuai/PhysHOI` `6095c605e2`	`physhoi/env/tasks/physhoi.py` `compute_humanoid_reward`	object-rotation weights 0.1 and 0.01 for GRAB, Table 4	`eor` and `eorv` set to `torch.zeros_like(ep)`, the computation commented out on the same lines
UniDexGrasp [150]	high	`PKU-EPIC/UniDexGrasp` `36c9bfcf7c`	`dexgrasp_policy/dexgrasp/tasks/shadow_hand_grasp.py` `compute_hand_reward, goal_cond branch`	seven named weights from 0.1 to 10, Table 7	a `torch.where` cascade on literals -0.5, -1.0, 0.9, 0.1, 0.2, none of the seven present
PenSpin [72]	medium	`HaozhiQi/penspin` `5035c52dc9`	`configs/task/AllegroHandHora.yaml` `forceScale`	a disturbance force of 0.2 times object mass at probability 0.25, Table 8	`forceScale: 0.0` and `randomForceProbScalar: 0.0`, the only shipped config naming either
PianoMime [31]	high	`sNiper-Qian/pianomime` `c4abefac8d`	`single_task/piano_with_shadow_hands_res.py` `_set_rewards`	two weighted terms, 2/3 and 1/3, Table 3	five summed terms; `_compute_energy_reward` ends `return 0` and `_compute_fingering_reward` `return 0.0`

papers whose robot has two end effectors, is not a count of bimanual dexterous manipulation at all. Thirteen of those put no dexterous hand on the robot: five are parallel-jaw throughout [127], [151], [152], [153], [154], four name no hand and run grippers on every embodiment [21], [22], [23], [128], two report every bimanual number on grippers and show five-fingered hands only qualitatively [144], [25], and two name no end effector anywhere [82], [28]. Three more run a gripper on one embodiment and a hand on another [121], [129], [120]. Appendix A lists every exclusion, so the denominator can be audited rather than taken on trust.

: the papers whose notes place a learned closed-loop controller on two multi-fingered hands. It is the 53 less those 16; less two static grasp-synthesis methods, BimanGrasp [41] and BiDexGrasp [155]; less three systems with no learned policy, Castro et al. and DexTeleop-0 [92], [68] and Pang et al. [35]; and less four papers where no learned policy holds both hands: OmniH2O [27], whose fingers are mapped open-loop from the Vision Pro and sit outside its 19-DoF policy; OKAMI [135], whose headline pipeline is open-loop retargeting with a learned policy only in a side experiment; DexDeform [32], a skill model refined by trajectory optimization rather than a closed-loop controller; and Omnigrasp [11], a simulated human body with no bimanual task. Frau-Alfaro et al. [20] never enter, because their entry records one hand. Benchmarks and datasets are outside the 28 by class, Bi-DexHands and Bench2Dex [46], [87] and RoboPianist [29] being benchmarks and RP1M [30] and HumanoidGen [156] datasets; they are quoted here as evidence and never counted. Every paper below runs two multi-fingered hands unless said otherwise.

A. Coupling, roles and collision

When both hands hold the same object, the object closes a kinematic loop between them. Neither hand can move without changing what the other must do. BiDexGrasp [155] reports that the coupled objective "often yields imbalanced solutions, where one hand dominates stability while the other contributes marginally." ArtiGrasp [157] reports simulation speed scaling "roughly quadratically with the number of contacts," and trains each hand alone before pairing them.

Role asymmetry is an assumption most of these systems make silently. BiDexHD [158] states it outright: "we assume the robot to be right-handed by default, i.e., the left hand handles the target object and the right hand handles the tool," and Lin et al. [159] bake the same split into their reward, putting reference contact keypoints on the bottle base for the left fingertips and on the lid for the right. The bias is in the data first: TACO [160] recruited 14 right-handed subjects and measures the right hand moving consistently faster.

The arms collide, and the corpus handles this by construction rather than by control. RoboPianist [29] ships a forearm-forearm collision term that its own reward table never lists. Two papers name arm collision as unsolved. DexMimicGen [45] "does not explicitly handle collisions," and DexMan [141] says its control parameterisation "overlooks full arm posture," which it needs to avoid them.

The action space doubles and the observation grows faster: a two-hand benchmark carries action vectors in the tens and observations in the hundreds of dimensions [46], and a recent generalist emits a chunk of arm joints, end-effector poses, hand joints and fingertips together [83]. What the larger vector does not carry is any statement of who does what. The usual answer is one scalar reward summed over both sides, and the cost is a

compounding failure rate: where success requires both hands to succeed, ManipTrans [17]'s rate falls by about a third from single-hand sequences to bimanual ones.

B. Coordination architectures

Fig. 10 sets the four mechanisms side by side. Every one of the 28 carries exactly one architecture, read from its note: one policy over both hands, a network per hand, leader and follower, or a note that does not say. The fourth panel is not a fourth architecture but the relative frame, and it is the same single paper as the third.

Twenty-one of the 28 put one policy over both hands, which is three quarters of the set: the trackers and tracking-adjacent methods [18], [141], [17], [140] and [26]; the reinforcement-learning tasks [109], [159], [15], [31] and [106]; the demonstration pipelines [44], [125], [161], [45], [50] and [126]; and the generalist policies [83], [145], [162], [130] and [163]. In every one of them a single network takes a concatenated two-hand observation and emits a two-hand action. Two papers do not say which they are, Bunny-VisionPro [43] and DexImit [148], whose notes describe the rig and the data pipeline but never the policy's own decomposition. The concentration is not the outcome of a comparison that was won.

Four of the 28 give each hand its own network, and the two papers that compare the choice disagree. Bi-DexHands [46], a benchmark entry and so outside the 28, ships the MARL baselines and finds PPO over the full observation beats HAPPO and MAPPO "in most cases," with the gap narrowing on tasks that need both hands, because PPO "can use all observations" where MARL sees only part. BiDexHD [158] concludes the opposite. Its independent PPO teachers score 74.59 percent stage-two tracking rate on trained tasks against 53.88 for a centralised policy over both observations. Both are simulation only, on different tasks and hands, so neither settles it. ArtiGrasp [157] trains one PPO policy per hand, and Dynamic Handover [19] and DyDexHandover [42] use MAPPO with one agent per arm-hand system.

One paper assigns explicit leader and follower roles, and it is the same one that expresses its policy in a relative frame. AsymDex [40] gives the dominant hand full finger and wrist control, restricts the facilitating hand to a base pose, and writes both the dominant hand and the object in a frame attached to the object the other hand holds, which halves the observation and the action. It is also the only paper here that ablates the two mechanisms separately, and neither works without the other: relative frames without asymmetry and asymmetry without relative frames both collapse to near zero where the full method succeeds most of the time. Its real rig pairs an Allegro with a cheaper hand because only one Allegro was available, so the roles are confounded with the hardware. Two other papers use a relative quantity without giving either hand a role, one appending the inter-hand pose to its observation [161] and one applying a shared transform to both arms while generating data [45], which is the mechanism offline rather than in a policy.

One architecture carries three quarters of the field, and the choice behind it has been compared twice, with opposite outcomes, and ablated once. A decisive comparison would run one handover task on three architectures with the same hand and the same seeds, reporting separate giver and receiver returns rather than a single shared one.

C. Benchmarks, datasets and the shared axes

Two purpose-built bimanual dexterous suites exist in the corpus, four years apart.

The two bimanual benchmarks in the corpus trade coverage against measurement, in opposite directions. Bi-DexHands [46] is 20 tasks on two Shadow Hands in Isaac Gym, ordered by the infant age at which humans acquire the skill, at 2048 environments and a reported 30,000-plus FPS. Its measurement discipline is weaker than its coverage. It reports reward and normalized score, never a success rate. Its only success flag in code tests the object-to-goal distance against 3 cm, which ignores orientation and exists only in the four catching tasks. Any success rate later work attributes to Bi-DexHands comes from that flag or its own definition.

Bench2Dex [87] is the more instrumented of the two, running 26 long-horizon tasks across 12 arm-and-hand embodiments with thousands of teleoperated demonstrations in eight modalities, including a ray-cast tactile image that maps every hand geometry into one representation. Its headline result is the gap between conditions rather than the leaderboard: the matched leader takes about half the rollouts and falls to a fifth under combined perturbation, which compresses the field until the ranking no longer holds. Tasks are not crossed with embodiments, so no contrast isolates the hand, and the authors state that their hand and tactile models are not calibrated against matching physical hardware.

Ten papers in this corpus are bimanual datasets, five of them two-hand interaction capture, and their annotations are where the contact problem first appears. ARCTIC [164] deliberately labels over-shooting vertices as in contact, so its ground truth records interpenetration as contact, and Fig. 11 is the chain that produces it. TACO [160] fits its sequences under an attraction loss and a penetration loss that the paper says trade against each other. OakInk-V2 [165] is the only one publishing a penetration figure for its own annotations, at a mean depth of a few millimetres. The two largest egocentric sets [166], [167] publish none at all. Those five feed most of the physics-based tracking work in this section [18], [157], [158], [17], [141], so an interpenetrating reference is where several of these pipelines start.

The method tabulation carries the bimanual papers on the same columns as every other method here, and three of those columns are worth reading together.

Real trials are few and are not counted the same way. Where a count is given it runs from ten per condition to a few hundred across an object set [40], [50], [45], [155], and one paper reaches real hands by open-loop replay with no trial count and no success rate at all [17]. Seven of the 28 have no real robot [157], [158], [18], [141], [109], [42], [31], and the two piano papers outside the denominator are simulation-only as well [29], [30].

Penetration, and Fig. 12 is why it has to be reported beside a success rate rather than instead of one. Most bimanual papers

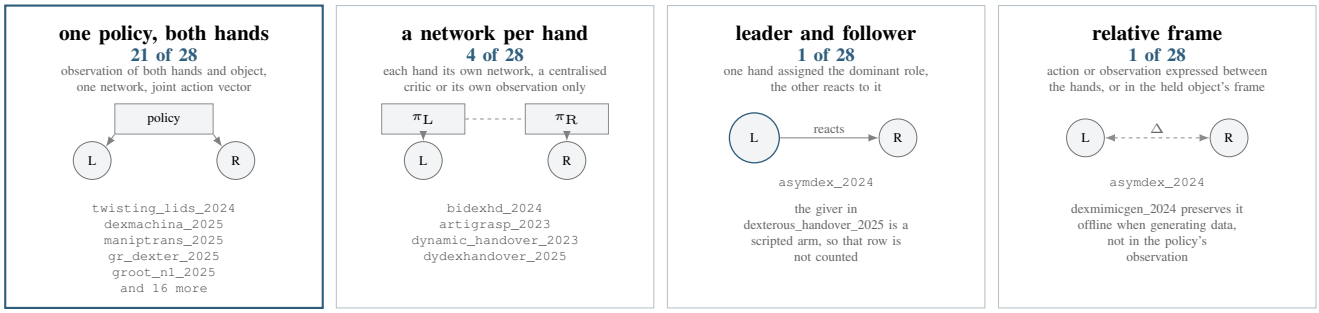
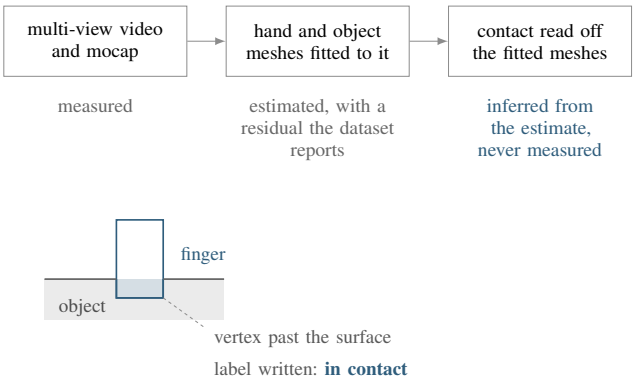


Fig. 10: Four mechanisms for controlling two dexterous hands, counted over the 28 papers in this corpus with a learned closed-loop controller on two multi-fingered hands, each read from its note. The panels are not a partition: the fourth is the same single paper as the third, and the two papers whose notes never say which architecture they use appear in neither.



The annotation is as good as the fit and no better, and a penetration is recorded as a contact rather than as an error. Of the five bimanual capture sets in Sec. VI, one publishes a penetration number for its own annotations.

Fig. 11: Where a capture set's contact annotation comes from. Nothing in the chain is measured at a surface, and a vertex that has gone through the surface is written down as a contact rather than as an error.

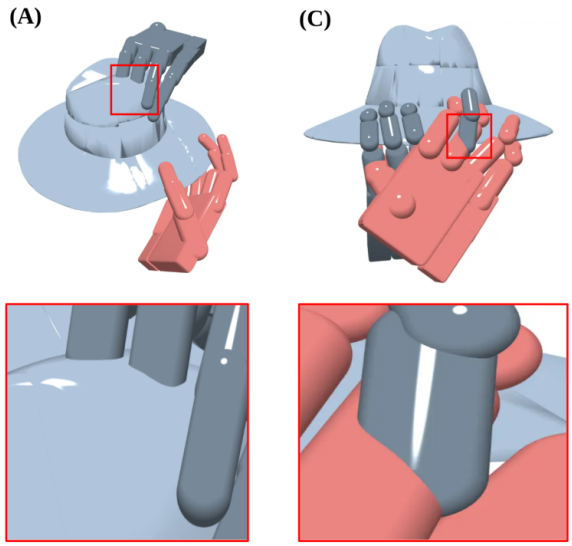


Fig. 12: Two failures of synthesised two-hand grasps, each enlarged at the contact: (A) a finger inside the object, (C) a finger inside the other hand. Both look plausible whole [41].

Fig. 9 of Shao and Xiao, arXiv:2411.15903, CC BY 4.0; panels (A) and (C) of Fig. 9.

read "not addressed." BimanGrasp [41] gates on it, failing any grasp whose total penetration across hand-object, self and inter-hand exceeds 1.5 mm, and reports that "penetration remains the primary cause of grasp failure." BiDexGrasp [155] reports it as a number, 0.15 to 0.20 cm maximum depth for its own grasps. DexImit [148] carries a named hand-hand penetration term in its grasp-synthesis objective. DexMachina [18] resolves it once, replaying retargeted joints against a fixed object "to eliminate object penetrations" and measuring nothing during rollout. ArtiGrasp [157] declines the metric outright, because its baselines "include a physics simulation which exhibits no interpenetration."

Comparability fails even on identical grasps. BimanGrasp [41] reports 54 percent success in Isaac Gym; BiDexGrasp [155] re-runs the same grasps in MuJoCo at a lower friction coefficient and gets half that, with more than a centimetre of penetration depth. Neither number transfers, and Section V-C's re-implementation result is the same lesson on the training side.

D. Handover, and what transfers from one hand

Handover is the one bimanual task where the hands are unambiguously asymmetric, because one gives and one receives. All three corpus handover papers use a single shared reward across giver and receiver. None defines separate objectives for the two roles.

Whether the two hands optimize the same objective is usually left implicit. Dynamic Handover [19] states one formula, $r = r_{\text{dis}} + r_{\text{linvel}} + r_{\text{torque}}$, and never says whether thrower and catcher receive different decompositions of it. DyDexHandover [42] is explicit that they do not: "Both hands share aligned objectives, forming a fully cooperative relationship," with one weighted sum feeding both agents. Only the thrower gets an extra KL regulariser toward a policy pretrained on human throws, which constrains style and not role.

The third case is weaker still. In [20] only the receiver is learned. The giver is a UR5e with an Allegro hand that "holds the object without moving during the whole episode." There is one agent, one reward and no second policy, its entry

accordingly records one hand, and it is one of the papers the 28 excludes. The 94 percent often attached to this paper needs its conditions. It is Total Success, which counts "Indetermination" cases where the simulator failed to resolve collisions and the object clipped through the giver's hand, on the short prism, in simulation, over 100 episodes, with no real robot in the paper.

Nothing in the handover literature measures contact quality. Contact appears only as a positive signal, a boolean per-phalange touch in [20] and a boolean contact reward in [42]. The two systems that do handover well sidestep the problem. HATO [50] reaches 10 of 10 on a real slippery handover by imitating teleoperation with fingertip touch sensing and no physics model. Lin et al. [106] stage their reward with a discrete variable switching which hand's fingertips are scored, and still calls handover its hardest task at 52.5 percent real success.

The training recipe transfers intact. PPO with thousands of parallel environments, an asymmetric critic on privileged state, domain randomisation and distillation into a vision policy is the same in [159] and [106] as in Section V-B. Reward forms transfer literally, and the contact-goal reward $r_{\text{contact}} = \sum 1/(1+\alpha d)$ appears in both.

Multi-task learning does not transfer, and the bimanual benchmarks show it clearest. Bi-DexHands [46] finds multi-task PPO and ProMP fail on MT4, MT20, ML4 and ML20. RoboPianist [29] finds F1 on the shared training song drops "from roughly 0.7 F1 for 1 song to almost 0 F1 for 16 songs," with no positive transfer "regardless of the size of the pre-training tasks."

Single-hand evaluation does not transfer either. Success criteria compound, because ManipTrans [17] fails a trajectory "if either hand fails to meet these conditions." And object-centric metrics cannot say which hand caused the failure. BiDexHD [158] scores the steps where both objects track their references, and DexMachina [18] averages ADD across object parts before an AUC. Both say whether the task worked and nothing about which hand lost it.

One failure mode has no single-hand counterpart. Two hands can penetrate each other, and the corpus almost never looks. Three papers carry an inter-hand penetration term, and all three are static grasp synthesis: [41], [155] and [148]. Not one learned bimanual controller in the corpus measures or penalizes hand-hand penetration during a rollout. PianoMime [31] ships a flag that turns inter-hand collision off in the physics.

VII. Evaluation, and a protocol for comparison

A. Reporting practice

Most of this literature does test on hardware, which is the check that matters and the one a simulator cannot substitute for: 89 of the 113 papers report a real-robot experiment. What they report about those experiments is much less even. Fig. 13 draws six reporting axes, each against the denominator that belongs to it, and two of the six are the ones a reader most needs. Without a count of unseen test objects there is no denominator for a claim about generalisation; without some statement about contact there is no way to tell whether the hand passed through the object. Those two are the axes the field states least often.

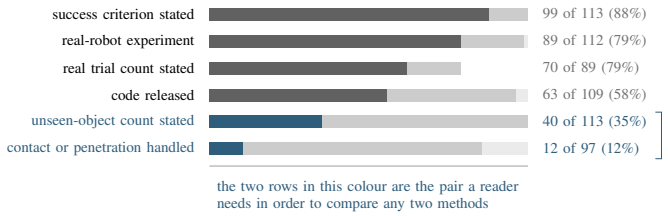


Fig. 13: What gets reported, and what does not. Each bar is a count of the 113 corpus method papers; the pale track behind it is the denominator that statistic is quoted against, printed beside the count, because four of the six are not 113. A faint tail past a track is the papers whose notes left that question unsettled. The two rarest reporting practices are marked in color.

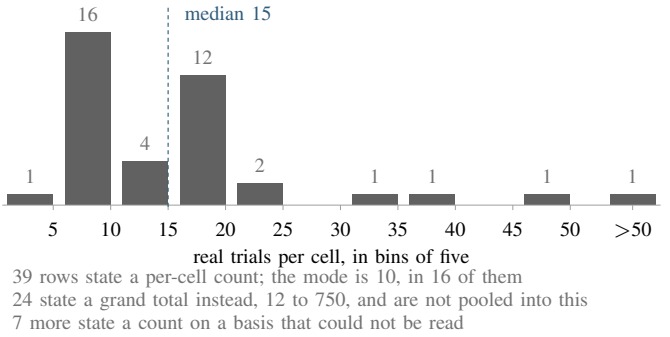


Fig. 14: Where a denominator is stated it is small: the 39 per-cell real-trial counts, binned by fives, median 15 marked. The 24 grand totals are a separate quantity, counted beneath the axis rather than pooled in.

Every share in the figure is a lower bound rather than a rate, for reasons Appendix A gives.

The plainest gap is the trial count. Nineteen of the papers that ran a real robot never say how many times they ran it, and a percentage with no denominator cannot be given an interval, so it cannot be compared with anything.

Where a count is stated it is small, and it is not always the same quantity. Some papers report a per-cell count, a fixed number of trials on each task or each condition; others report a grand total over every cell; a few cannot be assigned to either basis. The two distributions are kept apart here and in Fig. 14, because pooling them produces a figure that describes nothing: Hora [75]'s 240 trials and Visual Dexterity [16]'s 20 are experiments of comparable size recorded on different bases.

The per-cell count is the one a confidence interval attaches to, and at the sizes this field uses the interval is wider than the differences being claimed. Writing $\hat{p} = s/n$ for s successes in n trials and $z = z_{1-\alpha/2}$ for the normal quantile at the chosen level, the Wilson bounds are

$$\frac{2n\hat{p} + z^2 \;\pm\; z\sqrt{4n\hat{p}\,(1-\hat{p}) + z^2}}{2\,(n + z^2)}, \tag{1}$$

which, unlike the normal approximation, stays inside $[0, 1]$ and does not collapse to zero width at $\hat{p} = 0$ or $\hat{p} = 1$. At twenty trials per cell, two methods twenty points apart in success rate are not statistically distinguishable: the two-proportion test on that comparison returns nothing, and Appendix E gives the intervals and the test. The modal cell in this corpus is smaller

still. This is the single most consequential fact about evaluation in dexterous manipulation, and it is why Section VII-C proposes a protocol rather than a league table.

Reporting has improved a little and the experiments have not. About two thirds of papers from 2025 onward state a trial count, against three fifths before, but the counts themselves have not moved: the median has sat near twenty trials per cell every year since 2024, and earlier years rest on too few observations to read as a trend. A field can get better at disclosing a number without the number itself getting any better.

The denominators also sit on different hardware. The papers that name their own hand describe it in 79 different ways, unnormalised, so that figure counts strings and not designs; Fig. 3 gives the distribution. A success rate on an Allegro and a success rate on an Inspire hand are not measurements of the same thing, because the two differ in actuated joints by nearly a factor of three.

Nor are the criteria the same. DexVerse [88] counts a pick a success when the cube is "lifted at least 0.20 m above its resetting height". Bench2Dex [87] requires its terminal predicate to hold for a continuous dwell time, to reject transient contacts. The Colosseum [168] counts an episode successful "if the model completes the task fully". DexTrack [116] reports every success rate as a pair under two threshold sets, which can move the same rollouts by nearly twenty points. Among the papers that state no criterion at all, most have no success predicate to state: they report radians rotated or time-to-fall and never define a success, which is a fact about the paper rather than a gap in this survey. Where a criterion is stated it is usually a rubric, a staged partial credit or a deferral to a benchmark's own definition rather than a threshold, and a rubric is reproducible inside a lab and not across two.

Seven quantities dissociate in the published data, which is why they have to be reported separately rather than collapsed into one number.

Task success is the first, and binary success discards the difference between a near-miss and inaction. Snyder et al. [169] put it plainly: "a policy that completes 90% of the task is clearly better than a policy that is frozen the whole time, yet their success rates would be identically 0%". In [170] one policy scores in the teens overall on a pancake task while picking up the spatula and flipping the pancake in every attempt.

Robustness to perturbation has been measured, though not yet on a hand. The Colosseum [168] reports a success drop of a third to a half under single perturbation factors and more than three quarters under all of them together. In [87] the leader of the matched condition is not the leader under combined shift. The ranking at the anchor is not the ranking under shift.

Generalisation to unseen objects is stated by 40 papers, and half of those test on ten objects or fewer.

Physical plausibility of the contact is the least reported of the six, at 12 percent of the papers whose notes settle the question. Section VII-B takes it up.

Sample and wall-clock cost is reported by about a third of the corpus as an environment count and by fewer as an episode count. RoboPianist [29] is the exception that shows what disclosure looks like, giving samples per song and hours per run on named hardware.

Real-robot transfer is the one that undoes the rest, because simulated rank order is not real rank order. AutoEval [171] scores one policy at 6 of 50 in simulation and 47 of 50 on the real arm for the same task. Badithela et al. [172] state the limit directly: "the simulation-to-real gap precludes rigorous statistical inferences about real-world outcomes from simulation results alone".

Reproducibility is the last, and releasing code is not the same as releasing something that settles a question. Eight papers state a training objective in their own repository that differs from the one in the paper, which Section V-F and Table II set out file by file. That is the figure this section uses.

B. Physical plausibility

One quantity is specific to hands and is almost never reported: how far the hand passes into the object it is holding. Outside closed-loop control it has been a standard comparative column for years. Grasp synthesisers fail a grasp that penetrates past a stated depth [41], retargeting methods report a maximum depth against a baseline [143], and hand–object datasets score a split on depth, intersection volume and simulation displacement [173]. Among learned closed-loop policies almost nobody reports it, and nobody reports it measured in the physics the policy ran in.

D-Grasp [174] is the instructive case, because it does report a volume for a pose its own policy reached, and the way it does so shows where the quantity goes. Inside the simulator the paper reports zero penetration, since the meshes there are decimated for speed; the figure it prints instead comes from re-measuring a single frame against the original hand mesh and the full-resolution object mesh, with, in the paper's words, "no physical simulation involved". The number a reader wants, how deep the hand sits in the object in the physics that trained the policy, is zero by construction, and the number reported is a geometric re-measurement on geometry the simulator never used.

The same paper supplies the reason to want the measurement. Its appendix observes that a deeply penetrating reference grasp can raise a success rate rather than lower it, because "the objects can become entangled within the hand mesh and will therefore not be able to fall down", and concludes that "the success rate metric should always be interpreted in combination with the other metrics". A success rate with no plausibility number beside it can be read two ways.

There is a reason the gap persists, and it is not difficulty. A quantity a policy optimises cannot also judge it, because the policy learns the measure rather than the property the measure stands for. PhysHOI [142] shows the failure in the clean direction: its imitation reward is maximised by not touching the object at all, because contact perturbs the reference trajectory. DexTrack [116] shows it from the other side, defining a maximum penetration depth, applying it only to its input references, and then reporting tolerance of "severe hand-object penetrations" as a result. The remedy is to measure with something the policy never optimised.

Definitions are not shared either, which is the other half of why published figures do not compare. GRAB [175] reports an

average penetration in millimetres without saying whether it is a maximum, a mean or a median; OakInk [173] supplies the fullest published vocabulary, separating penetration depth, solid intersection volume and simulation displacement. A number without its definition is not a number a second paper can be held to.

What would close the gap is one line in a results table: the maximum and mean penetration depth over the evaluation rollouts, on a dense surface sample, computed by code that never entered the reward or the termination rule. That is the plausibility row of the protocol in Table III. What this survey supplies there is a measurement and a count; it does not supply a threshold, and Section VII-D says where the one in the table is borrowed from and why it does not bind.

C. Statistical practice, and a proposed protocol

Kress-Gazit et al. [170] give the field's reference protocol, and it prescribes process rather than numbers: write the success criteria before the run, have someone other than their author score the runs, match initial conditions with image overlays, interleave the policies blind within one session, report counts rather than percentages, and use a posterior over the success probability instead of a point estimate. All of that is right and none of it fixes the sample size. It prescribes no minimum trial count and no interval width, and its own worked example shows why one is needed: at twenty evaluations per policy, an eighteen-point difference in success rate still leaves better than a one-in-ten chance that the worse policy is the better one. Multiply every count in that example by ten and the same two rates separate.

The TRI LBM Team [176] supplies the missing numbers by fiat rather than derivation: 50 rollouts per task per policy per condition on hardware, 200 in simulation, blind, with randomised policy order inside per-initial-condition bundles. It replaces confidence intervals with Beta-posterior violins and corrects its pairwise tests under Bonferroni, "unless otherwise noted". Its own warning is the strongest sentence in this literature: "there is significant risk that many robotics papers are measuring statistical noise due to insufficient statistical power".

The rest fall short of their own advice. RoboArena [177] runs hundreds of double-blind pairwise comparisons across seven institutions and thousands of rollouts, and reports neither confidence intervals nor per-policy trial counts. The Colosseum [168] evaluates hundreds of test sets with "one training seed and one evaluation seed" per baseline and gives no intervals in simulation. AutoEval [171] calls ± 10 percent "the natural variance of robot evaluations" without giving the formula behind the intervals it plots.

Two papers do give usable numbers. Badithela et al. [172] pair real and simulated trials and de-bias the simulation with a rectifier, which saves up to a quarter of the hardware trials when the paired correlation is high and nothing at all when it is near zero; their decision rule is exact, that combining helps only when the rectifier variance is below the variance of the real evaluations. Snyder et al. [169] give the larger saving by scoring progress rather than success: a sequential test on graded scores cuts hardware trials by about 45 percent, and a gap that needs 80 binary trials to establish is established in 18 when the score is continuous. That is the strongest argument in this literature for grading partial progress.

Every count below is printed by `tools/make_eval_tables.py` in its derivation mode, and each axis is derived for the statistic that axis actually reports: a single rate takes a Wilson half-width, a matched comparison takes McNemar, a ratio takes the standard error of the log ratio, a correlation takes the Fisher-z interval.

Appendix E derives the counts in Table III, and they come to this. Reporting a single rate honestly takes about 100 real rollouts per cell, which is where the worst-case half-width of (1) falls to ten points. Comparing two policies is a harder question: the design is paired, so the count follows McNemar and depends on how often the two policies disagree rather than on the two rates alone, and separating a twenty-point difference at conventional power takes 57 matched pairs per arm, fewer rollouts than two independent arms would need. Simulated cells are cheap enough to run in the hundreds.

The other axes follow the same logic. Perturbation axes are screened cheaply, which ranks them but cannot establish that any one of them hurt, and certified at the cost of a single rate. Unseen-object claims resample the object rather than the trial, so the count that matters is the number of objects and not the number of rollouts. The transfer axis needs real and simulated trials paired, and separating two correlations takes several hundred pairs, which is why it belongs in a benchmark rather than in a paper. Every count there is a cap rather than a bill: a sequential test on a graded score can decide in a fraction of the trials [169], and a cell that stops early reports a confidence sequence rather than a fixed-sample interval, never both.

D. Feasibility and limitations

The proposed protocol has not been run. In the twelve-method comparison assembled for this survey, no method supplied all the measurements needed for a complete protocol row. DeXtreme [14], for example, reports a success criterion and a confidence interval, but uses ten trials. This limits retrospective comparison; it does not imply that the reported experiments failed their own stated objectives.

The protocol is expensive. For each policy and task it calls for 57 matched anchor trials and 100 unseen-object trials. Comparing two policies on three tasks is then 342 real rollouts on the matched set and 600 on the unseen-object set, 942 in all, or about 16 hours at one minute per rollout before repairs and scoring. The same bill under an independent-arm design is 1200 rollouts and 20 hours, so pairing saves about a fifth rather than the four fifths the pair counts suggest, because a pair is still two rollouts and only the anchor set is paired. AutoEval [171] completed about 850 WidowX episodes in 24 hours with three interventions and regular cooling pauses; a tendon-driven hand is likely to stretch the same workload across several days.

That bill is large but not unprecedented. Evaluating one generalist policy against its baselines has taken more than 2500 rollouts and more than 100 hours of human labour across three

TABLE III: The proposed evaluation protocol, and this survey's own contribution. No cell is empty: every cell is a proposal.

axis	what is measured	minimum trials	reported with it
Task success	fraction of episodes meeting a criterion written before the run, and a graded rubric score in $[0, 1]$	100 real and 200 sim per policy, task and condition for an absolute rate at a 10-point interval; a comparison alone needs 57 matched pairs per arm	raw counts, the criterion verbatim, the rubric, the initial-condition protocol, and an interval whose kind is named
Robustness	success under each perturbation axis, and the ratio to the unperturbed anchor	40 per axis per policy to rank the axes; 101 per axis and 101 at the anchor before any claim that a named axis hurt	both absolute rates with their Wilson intervals, the axis ranges, and the ratio without an interval unless the certifying count was run
Unseen objects	mean over held-out objects of per-object success	20 objects at 5 trials to screen; 93 objects for a 10-point claim	the object list, the per-object rates, a bootstrap interval over objects, and the within-object trial count
Physical plausibility	maximum and mean hand-object penetration depth per rollout, and the fraction of frames above 2 mm	100 rollouts, simulation only	an interval over rollouts, the sample density, the threshold, the solver's depenetration settings, and at least one rendered rollout
Sample and wall-clock cost	environment steps and wall-clock to the checkpoint that produced the headline number	3 training seeds	the range over seeds, and the statement that 3 seeds is a range and not an interval
Real-robot transfer	paired real and simulated outcomes on matched initial conditions, their correlation, and the rectifier variance against the real variance	the 100 real trials paired to 100 sim; about 457 pairs to separate two correlations 0.1 apart	the correlation with its Fisher-z interval, the variance ratio with its bootstrap interval, and which decision the count supports
Reproducibility	the released config and checkpoints against the paper's stated objective	not a trial count	the file and line of every disagreement, or an explicit statement that none was found

institutions [171], and a recent careful examination analysed close to two thousand real rollouts across nine hardware stations [176]. What is unprecedented is paying that cost for a single dexterous-hand paper, where the modal cell is ten trials.

Four things would have to change. Reviewers would have to reward 57 matched trials on one task over 20 unmatched trials on five, and nothing in the corpus suggests that is happening. The loop would have to be automated, and AutoEval [171] shows it is buildable for a gripper at one to three hours of setup, while stating that it supports binary success only and no robustness axes. Somebody would have to run the penetration measure on their own rollouts, which is a choice rather than a capability: Section IV-B shows the depth is computable from the poses and the meshes in a few lines of Warp, and IsaacGymEnvs already ships a task that does it every step. An earlier draft of this section made the engine the barrier, and that claim is set aside, because the released code refutes it. The comparison would also have to be sequential, because the savings in [169] are the only reason 100 is a cap rather than a cost.

The proposal has four limits. It has not been run, and its sample sizes come from power and interval calculations rather than dexterous-hand experiments. The worst-case calculation assumes a success rate of 0.5, while the paired design assumes a discordance rate of 0.3 that the literature does not report. Its perturbations come from parallel-jaw benchmarks [168], [178], where contact is simpler. Finally, the 2 mm penetration threshold follows [143]; it is a simulator convention, not a physical limit. Captured human grasps in GRAB [175] average 3.25 mm under the same kind of measurement. Establishing a meaningful threshold will require measurements across hands, engines, and accepted behaviors.

Seven papers in this corpus are evaluation protocols, a small denominator, and not one of them uses a dexterous hand: Badithela et al. [172] run a parallel-jaw gripper and the other six state no hand. Everything proposed above is therefore assembled from work on grippers and on whole-arm tasks. A modest first test would run the protocol in Table III once, on one in-hand reorientation task with one 16-DoF hand, and release the rollouts and per-trial outcomes for independent analysis.

VIII. Conclusion

This corpus describes a field that has converged. Most of it trains with reinforcement learning on privileged state in a GPU simulator and distills the result into a vision-only student; a second branch, growing faster, learns from teleoperated demonstration instead; and almost all of it runs on four hands. What the field cannot currently do is compare its own results, and most of the seven conclusions below follow from that.

1) Eight hands that can be bought today or built from published designs take zero papers between them, while the corpus's experiments concentrate on four designs (Section III-D, Fig. 3).
2) Human data does not port across hands, and the map is usually unstated: 33 of the 53 method papers that use human data never say how the human motion reached the robot hand (Section V-C).
3) Twenty-one of the 28 papers that put a learned closed-loop controller on two multi-fingered hands run one policy over a concatenated two-hand observation, a choice compared twice with opposite outcomes and ablated once (Section VI-B, Fig. 10).
4) The generalist and vision-language-action policies that do evaluate on a multi-fingered hand run it at a median of 6 actuated degrees of freedom, against 16 across the reinforcement-learning papers that state a count, and the papers reaching the larger band on paper never say whether the hand was in the evaluation (Section V-D).

5) The evaluation-methodology literature this survey builds its protocol on contains no dexterous hand at all: of its seven papers in this corpus, one runs a parallel-jaw gripper and the other six state no hand (Section VII-D, Table III).
6) Eight of the 63 papers that released parseable code state, in a named file at a named commit, something other than the value their paper prints. Table II gives each of the eight as a repository, a commit, a file and two values, and all of their authors were written to before this was posted: Section V-F says that beside the finding, with the letters in `outreach/` and the route by which a disputed case is corrected (Section V-F).
7) Penetration is almost never measured where it would matter. No paper here reports how far its hand sank into the object, measured in the physics its own policy ran in. The one that does report a volume for a pose its own policy reached measures it outside the simulator, on meshes the simulator never used, and reports zero inside it. On the reference-versus-rollout split that frames this survey, every paper that handles penetration at all is still measuring a reference (Section VII-B).

Six concern gaps in the literature. The first concerns publishing practice; the corrections made during this review are recorded beside the result in Section V-F.

Three of those claims carry a case worth remembering. A reward table in a paper is a claim about a document, not about a run, and PhysHOI [142] is the instance to keep in mind: the term its table weights at 0.1 is set to zero in the file at the commit this survey fetched, and the success criterion the paper reports is itself position-only. Contact is what separates a hand from a gripper, and the obstacle to measuring it is not the engines. IsaacGymEnvs ships a task that computes a per-environment maximum interpenetration depth against meshes and gates the policy update on a 1 mm threshold, and Tactile Genesis [179] offers penetration depth on Genesis geometry as a sensor. The tooling sits in the field's own benchmark repository and the number is still not reported. DexTrack [116] has the formula and points it at its inputs. And hardware and software have come apart on a narrower claim than the hand count first suggests, because 11 of the 19 hands that take no paper are neither sold nor open and take none for that reason, which is why the fifth claim is eight hands and not nineteen.

What this survey cannot establish is which method is better than which. It re-runs nothing, and Section VII argues that the published numbers do not compare. Six works are cited by metadata only, and a seventh, Ma and Dollar 2011, is on disk but unread; no claim rests on any of them. Every coverage figure here is a floor rather than a rate, for the reason Appendix A sets out.

Several practical changes require little additional computation. Authors can generate reward tables from the configurations used for the reported run, cite the exact revision, state the trial count and success criterion, and release per-trial outcomes. Evaluations can report penetration depth over policy rollouts as a diagnostic, without optimizing the policy against that measure. TopoRetarget [143] shows how informative the result can be. Hardware choices also deserve a broader base: the Allegro, Shadow, Inspire, and LEAP hands account for most of the published use in this corpus, while every other listed hand appears in eight method papers or fewer. Better reporting would make results easier to reproduce; broader hardware coverage would show which results survive a change of embodiment.

APPENDIX A
METHOD

The survey is written from a corpus that lives on disk. No claim about a paper is made from memory. Each claim traces to a note, each note to a parsed markdown file, and each markdown file to a PDF or repository with a recorded hash or commit.

Selection. Six topic-specific bibliographies were assembled in parallel (simulators and physics, hands and vendors, reinforcement learning, imitation and human data, bimanual, benchmarks and evaluation), seeded with the canonical works in each area and extended by search up to 2026-09-17. Every arXiv identifier was checked by fetching the abstract page and matching the title, and entries that could not be checked that way are marked. The six lists were merged with deduplication on arXiv identifier and normalized title, giving 222 entries.

One of those 222 was added after the search. `dgrasp_2022` entered the corpus on 2026-10-06, three weeks after the cutoff, and it turned out to be the clearest instance of the finding in Section VII-B. It was not hard to find: a corpus paper's own note cites it as that paper's baseline, and the search still did not surface it. A search that missed the work most relevant to one of this survey's two findings has probably missed others, and that is the plainest statement available of what this corpus is worth as a sample.

Seven further bibliography entries are not part of that corpus and are not counted anywhere in this survey. They are the prior audits and case studies Section I positions this survey against, they sit in a bibliography of their own as prior work from outside the corpus, and none of them carries a structured row. No source for them was parsed either, so each is quoted only from its abstract and its stated method.

Acquisition. PDFs were downloaded from arXiv or, for work without a preprint, from the publisher or vendor page recorded in the bibliography. Vendor pages for hands without any paper were fetched as HTML and converted to markdown, so that a specification quoted in this survey is quoted from a stored copy of the page and dated. Repositories were shallow-cloned and parsed, then deleted. What remains is one markdown per repository holding the README, a pruned file tree, the task and reward configuration files, and the bodies of reward and observation functions.

Parsing. PDFs were converted with pymupdf4llm, falling back to raw text extraction when the layout parse returned too little. The corpus manifest records, per entry, the source URL, page count, sha256 of the PDF and the converter used. Two consequences matter for reading this survey. Equations rendered as images do not survive conversion, so where a reward weight exists only inside a figure it is recorded as unreadable rather than guessed. Tables are flattened, so a number taken from a table is quoted with the table it came from and, where the flattening is ambiguous, the ambiguity is stated.

Notes. Each entry was read into a structured note under a fixed template: embodiment, learning method, objective, contact handling, evaluation, reproducibility, stated limitations, and quotable claims. Notes were written only from the parsed files. Where the source is silent the note says "not stated", and nothing was inferred from the reviewer's prior knowledge of the work. For method papers the reward or loss was quoted from the paper and, separately, from the released code, so that disagreements between the two are visible rather than smoothed over. Those disagreements turned out to be common enough to become a finding in their own right. The reading into notes was done with language-model assistance, from the parsed files only. The template's conventions are the safeguard against what that can get wrong: a quotation rather than a paraphrase, "not stated" rather than an inference, and a named source beside every value, so a note can be checked against the file it came from. Four fields carrying a headline claim were re-read against their sources afterwards, the trial counts, the success criteria, the unseen-object counts and the penetration field, and all four audits are below; the coverage statistics are reported as floors for the same reason.

Sources that could not be obtained. Six works are cited by metadata only and no claim in this survey rests on their contents. Five are behind publisher paywalls with no author-hosted copy found on 2026-09-18: Okamura et al. 2000, Bicchi 2000 (IEEE T-RO), Piazza et al. 2019, Roa and Suarez 2015, and Butterfass et al. 2001 on DLR-Hand II. The sixth is Hwangbo et al. 2018 on RaiSim, whose hosted PDF returned no body. The freely circulating PDF often taken for Bicchi 2000 is a different work, a book chapter, and is listed separately.

Ma and Dollar 2011 is a seventh case with a different cause. The fetch that failed when its note was written succeeded on 2026-09-18, so a seven-page PDF and its hash are in the manifest, but no note has been read from it. It is cited by metadata only for that reason and not because the source is unavailable.

Three of the 222 bibliography entries carry no structured row. Two are the paywalled Bicchi 2000 and DLR-Hand II entries. The third is [1], which was read into a note and quoted throughout but is a book chapter rather than a work with an embodiment, a method or a result to record in a row.

The reference list of this edition is shorter than the corpus, and the two numbers are different quantities. It prints 209 entries: the 202 corpus entries that some sentence, table or figure of this paper cites, plus the 7 prior-work entries from outside the corpus. The other 20 corpus entries carry a row and a note and are counted in every statistic here, but no passage in the paper names them, so they have nothing to be cited from and do not appear in the list. A reader counting the reference list should get 209, and a reader counting the corpus should get 222.

What this method cannot do. Mention counts over the corpus are counts of mentions, not of use: a related-work sentence counts the same as an experiment. Vendor specifications are manufacturer claims and are labeled as such throughout, and where a page has since gone offline the note says so. The corpus is large but not exhaustive, and selection by search favours work that is indexed, in English, and posted as a preprint. It also favours recent work. Of the 222 bibliography entries, 138 are dated 2024 or later and 16 predate 2018, so this is a corpus of the learned era and any claim here about a trend over time is a claim about 2022 onward.

Every coverage statistic in this survey measures what this survey's extraction captured, not what the literature reported. A structured row holds a scalar. A paper that reports a per-task count, a rubric, or a total spread across several tables produces a null, and a null is then counted as silence. The bias runs one way. Every miss converts a reporting paper into a silent one, so the field is made to look worse at reporting than it is. The size of the effect was measured on the statistic the survey leads with. Of the 34 method rows that had a real robot and no recorded trial count, 15 carried a count in plain text in their own note, dropped because the paper reports it per task and the field takes a single integer. Those 15 have since been re-extracted, which moved the stated-trial-count row from 55 to 70. The success criterion and the unseen-object count were audited the same way, rising from 79 to 98 and from 32 to 39. Both totals are one higher in the tables of this paper, because the row added after that audit states each of them. What those 19 recovered criteria say is mostly not a threshold: eleven score by rubric or staged partial credit, five judge binary completion against a task description by eye, two defer to a benchmark's own definition, and exactly one, [23], states a verbatim numeric threshold. The audit counted only cases where the note itself carried the number, so a count the note also missed is still uncounted, and 19 trial counts, 8 unseen-object evaluations whose object count the note never gives, and 4 criteria remain genuinely unsettled. Fifteen of the recovered trial counts are named in the rows: π_0 [21] at ten trials per task, RDT-1B [151] at 139 across seven tasks, UMI [154] at 260, $\pi^*_{0.6}$ [23] at 750, Gemini Robotics [144] at twenty per task, and ten more, which is 44 percent of the audited nulls and moved the "never says" figure from 34 of 89 down to 19. Read every coverage statistic in this survey as a floor rather than as a rate, and read the bars in Fig. 13 the same way.

The penetration field, audited the same way. The same mechanism reaches the penetration, code-release and failure-mode fields. The penetration field has since been audited by hand, because this survey's second finding is a null over it and a null is worth what the search behind it is worth. The population is the 85 method rows whose contact-handling field records that the work does not address penetration. It excludes the 16 nulls, which are already counted as unsettled rather than as silence, so recovering one would not move the finding. Twenty-five of the 85, 29 percent, were drawn with `random.Random(20260919).sample` over the sorted population, so the draw is fixed and can be redrawn. Each sampled row was read again in `papers/md/`, in its OCR recovery where one exists, and in `code/md/` where a repository was parsed — and not in `papers/notes/`, because the note is the artifact under suspicion: a field is wrong exactly when the source says something the note did not carry. A regular expression over the whole source collected every occurrence of penetration, interpenetration, intersection, intersection volume, solid intersection, simulation displacement, contact consistency, physical plausibility, signed distance and contact depth, and of

the words that are mistaken for them, and every hit was read in its context. A row counted as a recovery if its own source reported a measurement of any of those quantities on that method's own rollouts. A statement that penetration occurs, a citation whose title says physically plausible, a collision-avoidance constraint on a reference trajectory and a solver setting did not count, and the three recurring near misses of that kind are listed in Section VII-B. Recoveries: none of the 25, against the 20 to 45 percent the three earlier audits recovered on the fields above. The one-sided 95 percent bound, computed hypergeometrically over the finite population, is therefore 8 of the 85, and Section VII-B states it beside the claim and names the one borderline case that a reader might count differently. Why this field held where the others did not is visible in the per-row results: the recovered trial counts and criteria were numbers present in the source and dropped by a field shaped to hold a scalar, whereas a penetration number is absent from the source altogether. The earlier audits measured a defect in this schema; this one looked for an absence in the literature. `reviews/penetration_audit.md` holds the seed, the sample and a result per row beside the sentence it rests on, and `tools/audit_penetration.py` redraws the sample and re-runs the sweep. The code-release and failure-mode fields have still not been audited this way, and their counts stay floors.

Where the tabulation lives. Most of this survey's tabulation is not printed in this paper. The 113 method rows, the 21 reward-term objectives scored against their released code inside a 77-row extraction whose ten columns hold 253 cells no source stated, the 30 teleoperation systems, the 33 hand specifications with the source sentence behind each one, the 15 contact-model and solver descriptions verbatim, and the cell-by-cell mapping behind the predecessor-survey comparison: all of it is reference material, consulted a row at a time and never read through. Printed, it ran to more pages than the argument it supports. It is in the repository instead.

Read it there in one of three forms. `corpus/rows/` holds one JSON record per work, with every field at full length and no cell cut; this is the form to query, and it is the form every number in this paper is computed from at build time. `paper/tables/` holds the same content as markdown tables, one file per table, for reading rather than querying. `papers/notes/` holds the structured note each row was read from, with the quotation that supports each field, so that a cell can be traced to a sentence in a parsed source. The generators are in `tools/`: every table in this paper and in the repository is written by a script there from those records, so a reader who disagrees with a classification can change it and rebuild rather than argue with a PDF.

The corpus is not deposited yet, and this paper prints no link to it for that reason. It is available from the author at the address in the author block, and the archived release will carry a persistent identifier, which will be the copy to cite for any number quoted here. A survey that argues for checkable artifacts should point at its artifact rather than reprint it, and a table a reader cannot recompute is worth less than the record it was compressed from.

Three tables are exceptions and are printed, because the prose reads individual cells out of them and does not stand up without them: the two hand tables and the engine table, in Appendix B. The proposed evaluation protocol appears in the body alongside its justification. Table VII is the comparison to the fourteen predecessor surveys, which is this survey's positioning and is therefore set where a reader meets it rather than in the repository. It prints one of three levels per cell; Appendix D is the prose those levels coarsen, and the cell-by-cell mapping remains in the reproducibility materials.

Reproduced figures. One figure in this edition is not this survey's own artwork. It is a render reproduced from the paper it discusses, rendered from the stored PDF in the corpus, and it states its source in its own caption, keyed to the corpus entry it came from and to the page it was taken from. An earlier draft reproduced eighteen. A plate is kept only where it carries something a drawing cannot, and only where this paper has the right to print it: a contact defect invisible at the scale of a grasp and unmistakable at the scale of the contact, on a source that is CC BY 4.0. The other seventeen were redrawn in the paper's own style or dropped, and what each one was standing in for is said in the comment at the head of the figure that replaced it, or in the paragraph that absorbed it. Two of those seventeen went last and for a reason worth stating, because it is a limit on what this paper shows rather than a tidying of its figures. Both would have needed an author's permission, since arXiv's non-exclusive license to distribute grants a third party nothing, and this paper is posted without asking anyone for anything. One of them argued from geometry and was redrawn, as Fig. 8. The other photographed five research hands to scale beside a human hand and a ruler, and it was dropped rather than redrawn: the honest redraw is a to-scale outline annotated in millimetres, and no source in the corpus states a hand's length or its width, so every envelope in such a drawing would have been invented. Section III now states that absence instead of illustrating around it. The one surviving plate is cleared: its source carries CC BY 4.0, read directly from the license page on the date the register records, and its caption prints the attribution the license asks for, naming the creator, the work, the license with a link and the change made to the plate. No permission request and no author email is outstanding, and nothing in this paper waits on a reply. The register that records, per figure, the license found, the URL it was read from and the position it closed at, is `tex/PERMISSIONS.md`. The build does not enforce that register.

APPENDIX B
THE HARDWARE AND ENGINE TABLES

Sections III and IV argue over individual cells of three tables: which column is emptiest, which cell is blank and why, which figure rests on a press release. Those three are set here rather than in the body, because a reader consults them rather than reading them through. Every other tabulation in this survey is in the repository, and Appendix A says where.

Nothing in these tables is a measurement by this survey. Every specification is a manufacturer claim, a figure in a paper, or a figure in a press article. Table IV and Table V split the 33 hand rows by whether the hand can be bought

or built, and Table VI carries the 15 engines. The controlled vocabularies in all three compress a sentence the corpus stores in full: `corpus/rows/` holds the actuation, tactile, force-kind, contact-model and solver text at its own length, with the source's own punctuation and wording, and `papers/notes/` holds the quotation each was read from.

Each caption states how many of its own cells no source filled. Two conventions behind those counts are worth stating once. A blank price is a hand with no public list price rather than a free hand; and a hand with no entry in the manifest was recorded from a page that could not be stored, so its row should be read as a claim with no retrievable source.

Table IV holds the hands that can be had: sold, open-source, or with no release status stated. A row whose joint count no source states sorts last rather than being ranked by a number nobody stated. Its empty-cell count covers the eight specification columns, DoF through price; the hand, maker, open-hardware, status and usage columns are left out of it because they are never blank and counting them would dilute the finding. *force N* is not a ranking: read *force kind* first, because pull-out resistance, pinch force, a fingertip normal force under an indenter and an unlabelled vendor number are different measurements. *uses* counts the method rows whose own experiments run on that hand.

Table V carries the columns of Table IV plus the date of the claim and the class of artefact it was read off, over the same eight specification columns. Its empty share is far higher, and that is the finding of the table rather than a defect of it: an unreleased hand is announced without a specification. *evidence* is *datasheet* for a paper, datasheet or vendor specification page, *video* for a demonstration reel, *press* for a press release or a third-party tracker. *claim date* prints *undated* where the source page carries no date of its own; a fetch date is not a claim date.

Table VI prints, for each engine, the contact formulation it names, whether it exposes interpenetration depth and how many first-party hand models it ships; its empty-cell count is over the seven property columns. *contact model* and *solver* are controlled vocabularies: *soft* for a compliant or regularised contact, *convex relaxation* for a convexified formulation, *complementarity* for an LCP, *hard NCP* for a nonlinear complementarity problem, *impulse* for a velocity-level update, and *rigid, model unstated* where the source describes collision geometry but names no contact law. The sentence each was read from runs to several hundred characters; `corpus/rows/` carries each one verbatim, because Section IV's argument turns on its wording. *penetration exposed* is whether the engine makes interpenetration depth readable by a user's code without patching it. *hands* counts first-party hand models in the parsed snapshot, and they are: *Brax* synthetic 4-fingered claw hand; *Genesis* Shadow Hand; *Isaac Gym* Shadow Dexterous Hand, Allegro Hand, TriFinger; *Isaac Lab* KUKA Allegro hand, ShadowHand, AbilityHand, Fourier GR1-T2 dex hands, Unitree Inspire hand, Unitree trihand; *ManiSkill3* Allegro Hand, Ability Hand, Inspire Hand, Delto 3-finger hand, TriFinger; *Newton* Allegro Hand; *Orbit* Allegro hand.

APPENDIX C
PAPER AGAINST RELEASED CODE, IN FULL

Each entry below is the disagreement text stored in the row, unedited, and the class beside it is the one Section V-F and Section VIII count. A *contradiction* is a paper stating one value where the shipped code demonstrably states another. A *parse-limitation* is a comparison withdrawn because this survey's own parse could not settle it. *Code-absent* means the described component is not in the released repository, *version-skew* that the repository is a later generation than the paper, and *internal-inconsistency* that the paper disagrees with itself with no code implicated. Where a comparison was narrowed or withdrawn under review, the review note is printed beneath it. Every *contradiction* entry also carries an *Artifact* line: the repository, the commit `corpus/code_manifest.json` records, the file inside it and where in that file the value sits, so the entry can be settled without asking anyone. All of those authors were written to before this survey was posted. Section V-F says so beside the finding, sets out what a fetched commit can and cannot show, and states the route by which a disputed entry is corrected; the letters are in `outreach/` in the corpus, which is available from the author at the address in the author block. The reward-term extraction these rows were read out of is in the repository rather than here, with the rest of the tabulation Appendix A describes.

contradiction, 9 rows.

- PDDM [107] (high). Table 2 states obs-dim 46 for In-hand Reorientation while the released cube_env.py code sums to 39; the Baoding reward code includes an extra -10*wrist_too_high term absent from Table 2's printed formula.
  Artifact: `google-research/pddm` at `06b88cdbaf`, `pddm/envs/cube/cube_env.py` (`_get_obs`).
- DexPoint [37] (high). The released code's reward adds several terms absent from the paper's four-term Eq. 5 (a lift-threshold bonus, a target-distance term, a rotation bonus, and an IK controller-tracking penalty) and reshapes the reach/lift terms into inverse-distance and clipped forms rather than the paper's plain distance/height-difference formulas.
  Artifact: `yzqin/dexpoint-release` at `17f1e238bb`, `dexpoint/env/rl_env/relocate_env.py` (`AllegroRelocateRLEnv.get_reward`).
- DeXtreme [14] (high). Action Delta Penalty weight is -0.25 in paper Table 2 but -0.2 in the ADR yaml and -0.01 in the ManualDR yaml; the Joint Velocity Penalty in code normalises velocity by (max_velocity-vel_tolerance) unlike the paper's stated formula; code has a timeout_ rew term absent from the paper's reward table; Appendix Table 12 states critic learning rate 5e-4 and KL threshold 0.16, vs body text/code values of 5e-5 and 0.016.
  Artifact: `isaac-sim/IsaacGymEnvs` at `aeed298638`, `isaacgymenvs/cfg/task/AllegroHandDextremeADR.yaml` (`actionDeltaPenaltyScale`).
- Visual Dexterity [16] (high). Table S1 states 32000

TABLE IV: Hands that can be obtained, sorted by joint count. 68 of 160 specification cells are empty.

hand	maker	DoF	act. DoF	actuation	mass g	force N	force kind	tactile	price USD	open HW	status	uses
Shadow Dexterous Hand [51]	Shadow Robot Co.	24	20	tendon	4 300	–	–	optional	–	no	sold	21
Proception ProHand 1.0 [86]	Proception Inc	22	–	tendon	–	–	–	unstated	–	no	sold	0
SharpaWave 22-DoF [80]	Sharpa Robotics	22	22	–	1 300	20 / 12	grip	unstated	–	no	sold	6
BiDexHand [59]	Northwestern Univ.	21	16	tendon	–	2.14	fingertip normal	none	–	yes	open-source	0
Ruka-v2 [57]	NYU and NYU Shanghai	20	16	tendon	–	–	–	optional	1 500	yes	open-source	0
TESOLLO DG-5F [85]	Tesollo Inc.	20	20	direct drive	1 763	–	–	none	–	no	sold	0
Unitree Dex5-1 [84]	Unitree Robotics	20	16	tendon	1 100	10	fingertip normal	array	–	no	sold	0
Wuji Hand [81]	Wuji Tech.	20	20	direct drive	–	–	–	fingertip	–	no	sold	2
ORCA [60]	ETH Zurich	17	17	tendon	1 200	–	–	fingertip	–	yes	open-source	0
Allegro Hand V4 [61]	Wonik Robotics Co. Ltd.; earlier versions…	16	16	–	–	–	–	none	–	no	sold	35
LEAP Hand [58]	CMU	16	16	direct drive	595	19.5	pull-out	none	2 000	yes	open-source	12
RUKA [62]	NYU	15	11	tendon	–	2.74	pinch	none	1 300	yes	open-source	0
Inspire Robots RH56DFX [52]	Beijing Inspire Robots Tech. Co., Ltd.	12	6	–	540	10	vendor, no protocol	none	–	no	sold	19
ROBOTERA XHAND1 [180]	ROBOTERA	12	12	geared	1 100	–	–	unstated	14 000	no	sold	8
DEX-EE [181]	Shadow Robot Co., in collaboration with…	12	–	–	4 100	–	–	unstated	–	no	sold	1
BrainCo Revo 2 [182]	BrainCo Inc.	11	6	–	383	15	pinch	unstated	–	no	sold	1
AgiBot OmniHand 2025 [183]	–	–	–	–	–	–	–	–	–	–	unstated	1
DexHand [63]	IoT Design Shop	–	–	tendon	–	–	–	none	300	yes	open-source	0
LinkerBot Linker Hand [184]	LinkerBot	–	–	–	–	–	–	unstated	–	no	sold	1
PSYONIC Ability Hand [185]	PSYONIC	–	6	linkage	–	–	–	unstated	–	no	sold	7

TABLE V: Hands announced, prototyped or held internally, which cannot be obtained. 60 of 104 specification cells are empty.

hand	maker	DoF	act. DoF	actuation	mass g	force N	force kind	tactile	price USD	open HW	status	uses	claim date	evidence
Clone Robotics [78]	Clone Robotics	27	–	fluidic muscle	<910	–	–	array	–	no	internal-only	0	2025-11-24	video
1X NEO [76]	1X Tech.	25	25	tendon	–	45	vendor, no protocol	unstated	–	no	announced	0	2026-07-09	datasheet
LEAP Hand V2 [186]	CMU	21	17	tendon	–	–	–	none	3 000	yes	announced	2	undated	datasheet
Sanctuary AI Phoenix [187]	Sanctuary AI	21	–	fluidic muscle	–	–	–	unstated	–	no	internal-only	0	2024-12-17	press
ILDA hand [55]	Korea Univ.	20	15	linkage	1 100	34	fingertip normal	array	–	no	prototype	0	2021-11-05	datasheet
Pisa/IIT SoftHand [54]	Centro E. Piaggio, Univ. Pisa and IIT	19	1	tendon	–	–	–	none	–	no	prototype	0	2014	datasheet
Faive Hand [56]	ETH Zurich	16	11	tendon	1 100	–	–	none	–	–	prototype	1	2023	datasheet
Boston Dynamics [188]	Boston Dynamics	–	–	geared	–	–	–	unstated	–	no	prototype	0	2026-01-05	press
Daxo Robotics Muscle [53]	Daxo Robotics	–	–	tendon	750	–	–	–	–	–	prototype	0	2026-09-16	press
Figure 03 [79]	Figure AI	–	–	–	–	–	–	unstated	–	no	prototype	0	undated	press
PaXini DexH13 [189]	PaXini Tech	–	–	–	–	–	–	array	–	–	announced	0	2026-01-07	press
Tesla Optimus [77]	Tesla	–	–	tendon	–	–	–	unstated	–	no	prototype	0	2023-12-13	video
Xiaomi [190]	Xiaomi	–	–	tendon	–	–	–	unstated	–	no	prototype	0	2026-03-30	press

TABLE VI: 15 simulators and physics engines, by contact formulation and penetration exposure. 40 of 105 property cells are empty.

engine	contact model	solver	diff.	GPU	dt (s)	penetration exposed	hands
Brax [191]	convex relaxation	explicit/spring	yes	yes	–	no	1
ComFree-Sim [95]	complementarity-free	closed form	–	yes	0.002	yes	–
Dojo [94]	hard NCP	interior point	yes	no	–	yes	–
Genesis [98]	soft	Newton	yes	yes	–	–	1
Isaac Gym [91]	convex relaxation	PGS	no	yes	0.008333	no	3
Isaac Lab [96]	rigid, model unstated	Gauss-Seidel	no	yes	–	no	4
ManiSkill3 [102]	convex relaxation	PGS	–	yes	–	–	5
MuJoCo [100]	complementarity	Newton	no	no	–	–	–
MuJoCo convex [93]	complementarity-free	closed form	no	no	0.01	–	–
MuJoCo Warp [97]	soft	PGS	no	yes	–	–	–
Newton [99]	soft	primal-dual	–	yes	–	–	1
Orbit [101]	convex relaxation	PGS	no	yes	–	no	1
PyBullet [192]	–	–	–	yes	–	–	–
RaiSim [193]	–	–	–	–	–	–	–
SAPIEN [194]	convex relaxation	PGS	–	no	–	–	–

teacher training environments, but the released config sets alg.num_envs to 8000 (parent config 16384); fallDistance differs across two shipped configs (0.24 vs 0.15, only the latter matching Table S1's threshold); the paper's Eq 8 penultimate-joint penalty (c7=-2) does not appear anywhere in the released reward code; the config carries a dead distRewardScale=-10.0 key never used in compute_reward; and the paper's table-friction lower bound (0.05) differs by a factor of 10 from the code's randomized lower bound (0.005).
Artifact: `Improbable-AI/dexenv` at `ad9634e9d2`, `dexenv/conf/dclaw.yaml` (`alg.num_envs`).

- DexPBT [109] (high). Paper presents the reward as 4 mutually exclusive stage terms (r_reach, r_pick, r_targ, -r_vel), but code's compute_kuka_reward sums 8 named components, one of which (hand_delta_penalty) is multiplied by 0 and disabled; there is no single r_vel term in code, instead separate kuka/allegro action penalties whose exact formula is not shown. Also, the paper reports zero experiments with domain randomization, yet the shipped AllegroKuka.yaml already carries a fully specified DR schedule (disabled via randomize: False).
  Artifact: `NVIDIA-Omniverse/IsaacGymEnvs` at `aeed298638`, `isaacgymenvs/tasks/allegro_kuka/allegro_kuka_base.py` (`compute_kuka_reward`).
  Review: R3 adversarial review: the disabled-randomisation half is withdrawn, since the note finds it consistent with the paper; the zeroed reward term stands
- PhysHOI [142] (high). The released compute_humanoid_reward sets the body position-velocity error and the object rotation and rotation-velocity errors to zeros_like, unconditionally, with the computation that would produce them commented out on the same lines, while Table 4 lists nonzero λ^{or}=0.1/λ^{orv}=0.01 weights for GRAB: in that file the object's orientation error is the constant zero and those weights cannot change the reward.
  Artifact: `wyhuai/PhysHOI` at `6095c605e2`, `physhoi/env/tasks/physhoi.py` (`compute_humanoid_reward`).
- UniDexGrasp [150] (high). The paper describes a four-term weighted reward (r_goal + r_reach + r_lift + r_move via Table 7's omega weights) but the released compute_hand_reward implements a different threshold-gated torch.where cascade with distinct hardcoded coefficients that do not map one-to-one onto the paper's weights.
  Artifact: `PKU-EPIC/UniDexGrasp` at `36c9bfcf7c`, `dexgrasp_policy/dexgrasp/tasks/shadow_hand_grasp.py` (`compute_hand_reward`, `goal_cond branch`).
- PenSpin [72] (medium). The appendix states a randomised disturbance force and the released task config sets its scale to zero. A second half of the original comparison, that the released code disables the paper's tactile observation channel, is withdrawn: the config read has 96 observation dimensions and `enable_tactile: False`, consistent with the proprioception-only student policy rather than the tactile-and-point-cloud oracle, and the paper never claims the student has tactile input. The code's reward scale-key names (e.g. rotate_reward_scale, pencil_z_dist_penalty_scale) also do not 1:1 name-match the paper's Table 4 weight list, though matched values agree.
  Artifact: `HaozhiQi/penspin` at `5035c52dc9`, `configs/task/AllegroHandHora.yaml` (`forceScale`).
  Review: Narrowed while the letter to its authors was being drafted, before that letter went out on 19 September 2026. The tactile-channel half is withdrawn as a plausible reading of which pipeline stage the config belongs to; the disturbance-force half stands. Held at medium confidence pending a direct code read.
- PianoMime [31] (high). Paper's Table 3 states 2 weighted reward terms (Key Press 2/3, Mimic 1/3), but the released code sums roughly 5 unweighted terms (key press doubled, sustain, energy and fingering hardcoded to return 0, forearm-collision) plus a separately-added mimic wrapper term.
  Artifact: `sNiper-Qian/pianomime` at `c4abefac8d`, `single_task/piano_with_`

`shadow_hands_res.py` (`_set_rewards`).

internal-inconsistency, 4 rows.

- ALOHA [127] (high). Algorithm 1 pseudocode states L_ reconst = MSE, but Sec.IV.C's prose explicitly states L1 loss is used instead; the algorithm box and the implementation text disagree; code/md captures only function signatures, not bodies, for policy.py's loss implementation
- 3D Diffusion Policy [129] (medium). the paper's prose states the network predicts the noise added to the data, but the shipped default config trains with prediction_type: sample (predicting the denoised action a^0 directly), not epsilon; the paper only qualifies this later in the same section (Fig. 7 ablates both).
- OmniH2O [27] (medium). stumble weight -0.00125 (paper) vs -1250 (code); max-feet-height sign/magnitude differ (+1000 paper vs -2500 code, a penalty not a bonus); paper's exp(-c*//.//) form vs code's exp(-err^2/sigma); curriculum level-down threshold 40 (paper) vs 50 (code).
  Review: Withdrawn as a contradiction at the point of drafting the letter to its authors, which was never sent. Four of the five weight comparisons match the paper's table to the digit once a systematic $\times 1.25$ curriculum factor is accounted for, and only the stumble weight differs, by a factor of about a million, which is a typo signature in the paper's own table rather than evidence of a different trained objective. The hands in this work are also driven open-loop from VR pose, outside the policy and outside the reward, making it a poor fit for a dexterous-manipulation reward census regardless.
- DexMachina [18] (low). paper describes a plain weighted sum lambda_task*r_task + lambda_imi*r_imi + lambda_bc*r_bc + lambda_con*r_con with unspecified weights; code implements a multiplicative task term with per-component beta decay, an unmentioned 0.1 force-penalty term, and curriculum-driven decay of auxiliary weights not described as such in the paper
  Review: R3 adversarial review: the multiplicative form credited to the code is printed in the paper itself

version-skew, 4 rows.

- Hora [75] (high). The parsed code commit's joint-noise range, disabled default disturbance force, and default cube object type differ from the paper's stated values; the code README itself says to use tag v0.0.1, not the parsed commit, to reproduce paper numbers.
- π_0 [21] (high). The released repo (openpi, commit 215abfb2) has evolved past this paper: it also documents pi0-FAST (autoregressive) and pi0.5 checkpoints/configs not described in this paper, which describes only the flow-matching pi0 and the non-VLM pi0-small ablation.
- GR00T N1.6 [28] (high). code/md is the current Isaac-GR00T main branch (N1.7 generation, commit 51d4c89f), not the N1.6 checkpoint; the repo names a separate n1d6 branch for N1.6 that was not fetched. The page states N1.6's backbone is "an internal NVIDIA Cosmos-2B VLM variant," but the repo's own changelog states the Eagle backbone (nvidia/Eagle-Block2A-2B-v2) was used through N1.6 and only replaced by Cosmos-Reason2-2B in N1.7 – whether the page's Cosmos-2B variant is a third distinct model or the page is describing N1.7 under an N1.6 headline is not resolvable from what was fetched.
  Review: The page attributes a Cosmos backbone to N1.6 while the repository changelog says N1.6 used Eagle and N1.7 made the switch, so the repository is a later generation than the page describes.
- GR00T N1 [145] (high). The parsed code repo (Isaac-GR00T, commit 51d4c89f) is a later N1.7 generation, not the paper's GR00T-N1-2B: VLM backbone changed from the paper's Eagle-2 to Cosmos-Reason2-2B (via Qwen3-VL); action horizon expanded from the paper's stated H=16 to 40; state/action dimensions expanded from 29 to 132; DiT layers changed from 32 to 16; embodiment scope widened to include Unitree G1, AgiBot G1, and a 'YAM' arm not described in the paper.

code-absent, 8 rows.

- AsymDex [40] (medium). Code contains a GraspAndPlace-task reward variant (AllegroHandDualArmGraspAndPlaceAsymDex.py) with exponential hand-distance and block-alignment terms not one of the four BiDexHands tasks described in the paper body.
- DexMimicGen [45] (high). The environment code includes an unused 'reward_shaping' branch (if self.reward_shaping: pass) that is never described or used in the paper text; only the sparse binary completion reward is used, as the paper states.
- DrEureka [117] (high). The released DrEureka repo (code/md) contains only forward_locomotion/ and globe_walking/ trees; the LEAP-hand cube-rotation training/deployment code and the quoted Prompt-13 reward are the LEAP-hand authors' own code, not included in the released repo, so the cube-rotation reward cannot be cross-checked against a code file.
- Open-TeleVision [120] (low). The paper's prose states 25k training iterations, lr 5e-5, batch size 45, but the code repo's example training command instead shows 50000 epochs and an explicit kl_weight=10 not mentioned anywhere in the paper text.
  Review: R3 adversarial review: a behavior-cloning paper with no reward; it compared paper iterations to a README example's epochs
- Being-H0 [195] (high). The actual downstream action-head module referenced in §4.3 (learnable query tokens, proprioceptive projector f_p, regression head f_r) is not present in the parsed code; the repo's own TODO list marks 'Training code and scripts' as unreleased, so the action-chunk-length=16 value and exact proprioception-vector fields come only from inference/eval CLI examples, not a verifiable implementation.
- $\pi_{0.5}$ [22] (high). The paper describes a hybrid discrete-FAST-token-pretrain-then-flow-matching-post-train recipe (Eq. 1, alpha: 0 to 10), but the released repo states plainly it 'currently only support[s] the flow matching head for both pi0.5 training and inference.'
- $\pi^*_{0.6}$ [23] (high). The general openpi framework (pi0/pi0.5/FAST, flow-matching action head,

ALOHA/DROID/LIBERO adapters) is released at commit 215abfb2, but the RECAP-specific pieces described in the paper – the advantage-conditioning input, the distributional value function, the RECAP training loop (Algorithm 1), and the espresso/laundry/box-assembly checkpoints and datasets – are not present in the parsed repo.

- Being-H0.5 [196] (high). Released code (commit e12ac44f) exposes only benchmark/inference/config scaffolding (policy.py, beingh_policy.py, dataset-transform configs). no MoT/MoF model definition, no rectified-flow action-expert source, and no config for any dexterous-hand embodiment; the only post-train configs present (libero_robocasa.yaml, libero_all.yaml, robocasa_human.yaml, so101_example.yaml) are all gripper-only, so the dexterous-hand claims cannot be cross-checked against released code.

parse-limitation, 14 rows.

- ArtiGrasp [157] (low). paper's two regulariser weights (w_rh=0.5, w_ro=0.2) correspond to five separate velocity penalties in code (-0.5,-0.2,-0.5,-0.5,-0.3); the fingertip weight 12.0 and lambda 5.0 in Table 6 do not appear in the yaml; reward weights differ between curriculum phase-1 and phase-2 configs though Table 6 reports only one set
  Review: R3 adversarial review: the reward body is C++ and absent from the parse; the weight split is the paper's documented curriculum
- Diffusion Policy [128] (high). the paper's core loss equations (DDPM training loss, EBM/InfoNCE, score-matching) are unrecovered images in the parsed text, and code/md is a signature-only API map, so compute_loss's body could not be verified from code; only the exact shipped noise-scheduler config (DDPMScheduler, 100 train steps) is confirmed.
- Eureka [15] (low). The released repo's default eureka/cfg/config.yaml ships iteration:1, sample:3 (a fast-test default), not the paper's reported experimental configuration of N=5 iterations $\times$ K=16 samples $\times$ 5 independent runs, and LLM-generated reward code itself is never checked into the repo (written at runtime to a gitignored outputs/ directory).
  Review: R3 adversarial review: the repo README documents the paper's defaults; the cited config block is a fast-test preset
- UniDexGrasp++ [12] (low). The paper's four named reward weights (0.5, 0.1, 2, 10) numerically match literals in the released compute_hand_reward, but the code never labels them with the paper's variable names, and the paper's own display equations were dropped from the PDF, so the correspondence is inferred rather than confirmed by either source.
- BiDexHD [158] (high). Table 5 weights (w_r, w_t, w1-w4, lambda_w=0.12, lambda_ft=0.48) cannot be matched to code reward lines because the code md truncates the reward function before stage-2/final sum; lambda_w and lambda_ft numerically match the code's grasp-condition thresholds (0.12, 0.12x4 fingers) rather than confirmed reward weights
- GraspXL [47] (low). The code splits the paper's single r_reg = -w_h//u_h//^{2} - w_o//u_o//^{2} into four separately-weighted terms (wrist vel, wrist qvel, object vel, object qvel), and the floating-phase object-velocity coefficient (-1.5) does not match the paper's stated w_o=0.1, so the regularization weighting cannot be reconciled 1:1 with Table 9.
  Review: R3 adversarial review: the environment source is not in the parse and the note says the reward cannot be verified
- HumanPlus [26] (medium). code implements a superset of legged_gym reward terms (lin_vel_z, ang_vel_xy, orientation, base_height, torques, dof_vel, dof_acc, action_rate, collision, termination, dof/torque limits, tracking_lin/ang_vel, feet_air_time, stumble, stand_still, feet_contact_forces, target_jt) beyond the 8 terms documented in the paper's Table 1
- Omnigrasp [11] (medium). The code's compute_pregrasp_reward_time hard-codes w_pos, w_rot = 0.9, 0.1, silently overriding the rwd_specs weight arguments passed into the function; separately, the paper's pre-grasp reward weights (w_hp, w_hr) are not resolvable from either the paper text or Appendix Table 7.
- ResDex [149] (medium). The base-policy and residual-policy task files implement different variants of compute_hand_reward (whether finger/hand-distance terms are included alongside goal_hand_rew/hand_up/bonus differs), a difference the paper's Sec 4.1/4.4 text does not describe, and the released configs' env count (4096 default) and iteration counts (2500/5000) do not match the paper's stated 11,000 envs and 5,000/20,000 iterations.
- DexGraspVLA [10] (high). Hardware-interfacing code is explicitly withheld ("Due to intellectual property constraints, we are unable to open-source the hardware-related code"); only function signatures for the controller loss (compute_loss, noise_assignment) are present in code/md, not their bodies, so the MSE/Immiscible-Diffusion loss cannot be verified against code.
- DexTrack [116] (low). The paper gives one canonical reward-weight table (Table 3), but the released repo's task-config YAMLs contain several divergent, unreconciled reward-coefficient sets across AllegroHandTracking/...TrackingGeneralist/other variants, and it is not identifiable from the parsed source which config produced the paper's headline Table 1 numbers.
- DexVLA [48] (high). code/md only carries class/function signatures, not bodies, for the diffusion head (modeling_scaledp.py) and no config file with the α weight was captured; the paper's diffusion loss and denoising process cannot be verified against code bodies. only the multi-head ScaleDP signature and a DDIM fp32 patch are available, so none of the paper's tables can be reproduced from what was parsed.
- ManipTrans [17] (low). paper's PPO learning rate (5e-4) and env count (4096) differ from the shipped README/yaml defaults (2e-4, 8192)

Review: R3 adversarial review: the cited values are a README override and an unused fallback; the repo's own config matches the paper

APPENDIX D
THE EXISTING SURVEYS, AND WHAT EACH COVERS

Fourteen corpus entries are themselves surveys or engine-comparison studies. Table VII in Section I sets them on one set of columns, three levels to a cell, recording what each work covers and not how well. This appendix is the prose behind those levels: each glyph in that table coarsens one of the sentences below. Section I states what this survey adds over them and does not repeat it here.

The survey of An et al. [197] is the closest in subject, covering imitation learning for multi-fingered hands by learning family, end-effector class and demonstration source. It gives reinforcement learning no taxonomy, treats bimanual work as a single "multi-agent" subsection, compares no simulator, and defines no evaluation metric. The survey of Welte et al. [198] finds only seven dexterous works that use interactive imitation learning and carries a fifteen-hand commercial table, with no bimanual section, no benchmark table and no contact modeling. The survey of Zhao et al. [199] supplies the standard sim-to-real split and predates GPU-parallel simulation. The survey of Firoozi et al. [200] has zero occurrences of bimanual, tactile or in-hand.

The survey of Bai et al. [36] spans all of manipulation across 212 pages. Its Sec. 4.3 on dexterous manipulation runs about 720 words, the third longest of its ten task subsections behind grasping and quadrupedal manipulation, which is a real treatment and not a passing mention. The difference is elsewhere. Its Sec. 1.2 lists dexterous manipulation among the topics that "existing surveys" cover from "narrower perspectives" and defers it to two of them.

The survey of Zhao et al. [201] is the most recent hand-centred one, with a 29-hand anatomy table and a task-by-paradigm taxonomy. Its Sec. IV-C names the reference-versus-rollout split that Section I credits it with, and its Sec. III-F is the one bimanual subsection, 16 cited works with no coordination analysis.

The review of Kaup et al. [202] is the predecessor closest to the engine comparison, and it reviews Brax, Chrono, Gazebo, MuJoCo, ODE, PhysX, PyBullet, Unity and Webots for reinforcement learning research. It scores each on documentation, model and environment creation, URDF and MJCF support, and readiness for multi-agent work. It runs no benchmark of its own and says so in its Sec. V, that implementing the same scenarios across nine engines "goes beyond the scope of this paper". Its running bodies are ant-and-humanoid RL benchmarks rather than hands, and it discusses no timestep, no friction model, no contact formulation and no penetration.

The oldest rows carry the cost of the corpus. Okamura et al. and Piazza et al. [203], [204], and Roa and Suárez [205], are behind publisher paywalls with no author-hosted copy found on 2026-09-18, and Bicchi [206] is in the same position with no row at all. Ma and Dollar [207] are a different case. The fetch that failed when its note was written succeeded afterwards, so a seven-page PDF is on disk with a recorded hash, and no note has been read from it. All five are cited by metadata only and nothing in this survey describes their contents. The open chapter [1] overlaps the paywalled Bicchi paper without being identical to it, so it is quoted in its own right.

A cell is a level and not a score: ✓ where the work gives the topic a structure of its own, a section, a table or a measurement it defines; ○ where the topic is there without one, a subsection with no analysis, a handful of mentions, or a metric named and never defined; × where it is absent or put outside the work's scope. Each level coarsens one sentence read from `papers/notes/<key>.md`, and it is printed only while the quotation behind that sentence is still in the note, so a glyph cannot outlive its evidence. Appendix D prints the prose those levels coarsen; the cell-by-cell mapping is in `paper/tables/table10_surveys.md`. Three of the fourteen sources could not be obtained and one is on disk and has never been read, which is why their cells are empty; they are grouped as such rather than scored, because this survey knows nothing of them beyond a title and a DOI. The *gaps named* column is uniform by design: every predecessor names gaps, and what none of them attaches to a gap is a measurement.

APPENDIX E
WHERE THE PROTOCOL'S COUNTS COME FROM

Every count in Table III is derived below, and each axis is derived for the statistic that axis actually reports rather than by one convention applied to all of them: a single rate takes a Wilson half-width, a matched comparison takes McNemar, a ratio takes the standard error of the log ratio, and a correlation takes the Fisher-z interval. Section VII-C states what these derivations conclude. This survey re-ran no method, so every count here rests on an interval width, a power calculation or another paper's measurement, and never on a measurement of our own.

Fix the width first, then read off the count. Take a 95 percent Wilson interval on a single reported rate, at the worst case of $\hat{p} = 0.5$. Writing the half-width of (1) as

$$w(n, \hat{p}) = \frac{z\sqrt{4n\hat{p}\,(1-\hat{p}) + z^2}}{2\,(n + z^2)}, \tag{2}$$

the count a target width w_0 demands is $n_{\min}(w_0) = \min\{\, n : w(n, 0.5) \le w_0 \,\}$, which gives

half-width w_0 (points)	20	15	10	5
$n_{\min}$ (trials)	21	39	93	381

Ten points is the coarsest width at which a single rate is worth printing, so the absolute-rate minimum is 93, rounded to 100. At 100 trials a reported 80 percent has an interval of 71 to 87 percent, and a reported 50 percent has 40 to 60. A comparison is a different question and a harder one: two rates each carrying ± 10 points do not resolve a 10-point difference between them, because the difference's standard error is larger by a factor of $\sqrt{2}$, so the width argument sets a floor on what is worth reporting and not on what can be compared.

For the A/B comparison the relevant calculation is power, and the design is paired. Table III matches initial conditions by

TABLE VII: What 14 predecessor surveys cover: ✓ covered, ○ partly, × not. 24 of 84 coverage cells are empty.

survey	yr	dexterous scope	taxonomy	bimanual	hardware	evaluation	gaps named
Surveys of the field							
Zhao et al. [201]	2026	✓	✓	○	✓	○	✓
An et al. [197]	2025	✓	○	○	○	○	✓
Bai et al. [36]	2025	○	✓	○	○	○	✓
Welte et al. [198]	2025	○	✓	○	✓	×	✓
Firoozi et al. [200]	2023	×	○	×	×	○	✓
Zhao et al. [199]	2020	×	✓	×	○	×	✓
Simulator and contact-model studies							
NVIDIA Isaac Sim [208]	2026	×	○	○	×	×	✓
Kaup et al. [202]	2024	×	✓	×	×	○	✓
Le Lidec et al. [34]	2023	×	✓	×	○	✓	✓
Erez et al. [89]	2015	×	×	×	○	✓	✓
Sources not obtained, or on disk and never read							
Piazza et al. [204]	2019	–	–	–	–	–	–
Roa and Suárez [205]	2015	–	–	–	–	–	–
Ma and Dollar [207]	2011	–	–	–	–	–	–
Okamura et al. [203]	2000	–	–	–	–	–	–

image overlay and interleaves the two policies in one session, so the unit is a matched pair and the count follows McNemar, which depends on the discordance rate. The share of initial conditions on which the two policies disagree, and not on the two rates alone. Write $\delta = |p_1 - p_2|$ for the gap between the two success rates, π_d for that discordance rate, $z_{1-\alpha/2} = 1.96$ at $\alpha = 0.05$ and $z_{1-\beta} = 0.84$ at 80 percent power; then the number of matched pairs is

$$n_{\text{pairs}} = \frac{\left[\, z_{1-\alpha/2}\sqrt{\pi_d} + z_{1-\beta}\sqrt{\pi_d - \delta^2}\,\right]^2}{\delta^2}. \tag{3}$$

To separate 50 from 70 percent, which is $\delta = 0.20$, at $\alpha = 0.05$ with 80 percent power, (3) asks per arm for

discordance π_d	0.2	0.3	0.4	0.5
n_{pairs}	37	57	77	96

The protocol assumes 0.3 and asks for 57, and states the sensitivity rather than hiding it, because 0.5 is the discordance the same two rates produce when the pairing buys nothing, and at that value the paired count returns to the 93 per arm an unpaired test would need. The saving from pairing is real but smaller than the pair counts suggest, since a pair costs two rollouts: 57 pairs is 114 rollouts against 186. An earlier version of this section quoted 93, 169 and 387 per arm for gaps of 20, 15 and 10 points, which are correct for independent arms and are the wrong test for this protocol. That 93 was also the same integer as the half-width calculation in the paragraph above, which is a coincidence of the worst-case arithmetic and not a second derivation of the same number.

One hundred is a cap and not a bill, because on a graded score a sequential test reached its decision in 12 to 36 paired hardware trials in [169]. At 30 rollouts a continuous score already carries a half-width of ± 0.36 standard deviations, which is why a graded score can stop where a binary one cannot. A cell that stops early does not report a Wilson interval. Optional stopping breaks the coverage of a fixed-n interval, which is the reason Snyder et al. [169] and Badithela et al. [172] use anytime-valid betting intervals rather than Wilson, so Table III asks a cell run to a fixed 100 for a Wilson interval and a cell stopped early for a confidence sequence, and never for both. Intervals are also marginal rather than simultaneous. For example, evaluating twelve methods on seven axes produces 84 intervals. At 84 independent 95 percent intervals, four excursions are expected by construction, so a paper comparing k policies on m tasks corrects its $k(k-1)/2$ pairwise tests to a global 95 percent level, as the TRI LBM Team [176] does, or says its intervals are not simultaneous.

Simulation is cheap, so simulated cells take 200 episodes, giving a 6.9-point half-width. Perturbation axes are screened rather than certified, and 40 per axis buys a 15-point half-width on each axis's own absolute rate, which is enough to rank the axes and pick the two worst for hardware. It is not enough for the ratio to the anchor that an earlier draft asked each cell to report. At 40 trials in each arm, a fall from a 0.50 anchor to 0.30 is a ratio of 0.60 with a 95 percent interval of 0.34 to 1.06, which contains 1: at the screening count you cannot establish that the perturbation hurt at all. Certifying that same drop takes 101 per arm at 80 percent power, or about 50 for an interval that merely excludes 1, so Table III now asks for absolute rates with their own intervals at 40, reports the ratio without an interval, and prescribes 101 before any claim that a named axis hurt.

For unseen objects the resampling unit is the object and not the trial, so 20 objects at 5 trials each gives 100 trials and an object-level half-width near 20 points. That 20 points is the Wilson width at $n = 20$ and it treats each object's outcome as a single Bernoulli draw, which the five within-object trials are not. It is the right order of magnitude and the assumption belongs in the cell. A 10-point claim about an object distribution needs about 93 objects. Seven of the 39 rows that state an unseen count reach that: 225 in [41], 241 in [149] and [150], 360 in [10], 500 in [132], 2029 in [209] and 503409 in [47]. For a continuous score the half-width is 1.96 standard deviations over the square root of the count, $1.96\, s/\sqrt{n}$, so 100 rollouts give ± 0.20 standard deviations, and the unit is the rollout because frames within one are correlated.

The transfer axis is the one where 100 is least defensible. On 100 matched pairs a measured correlation of 0.70 carries a Fisher-z interval of 0.58 to 0.79, a half-width of about 0.10 that 130 pairs would be needed to guarantee. That is enough to establish that a simulator tracks reality at all, and it is not enough to separate [172]'s useful regime from its marginal one, since those differ by about 0.11 in correlation. Both limits come within 0.05 of the estimate only at about 457 pairs. Table III states which of the two decisions each count supports rather than leaving a reader to assume the larger one.

REFERENCES

[1] Bicchi, "Robotic Grasping and Manipulation (book chapter)," in *RAMSETE: Articulated and Mobile Robotics, Springer LNCIS*. Springer, 2001.

[2] OpenAI *et al.*, "Learning Dexterous In-Hand Manipulation," *IJRR 2020*, 2018.

[3] Collberg and Proebsting, "Repeatability in Computer Systems Research," *Communications of the ACM*, vol. 59, no. 3, 2016, related work outside this survey's corpus; no source parsed, no structured row, and no count rests on it.

[4] Raff, "A Step Toward Quantifying Independently Reproducible Machine Learning Research," in *NeurIPS*, 2019, related work outside this survey's corpus; no source parsed, no structured row, and no count rests on it.

[5] Engstrom, Ilyas, Santurkar, Tsipras, Janoos, Rudolph, and Madry, "Implementation Matters in Deep Policy Gradients: A Case Study on PPO and TRPO," in *ICLR*, 2020, related work outside this survey's corpus; no source parsed, no structured row, and no count rests on it.

[6] Knox, Allievi, Banzhaf, Schmitt, and Stone, "Reward (Mis)design for Autonomous Driving," *Artificial Intelligence*, vol. 316, 2023, related work outside this survey's corpus; no source parsed, no structured row, and no count rests on it.

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